Master Degree Project

UNIVERSITY OF SKÖVDE

I CAST CONTROL CHAIN OF THOUGHT

A prompt introduction of roleplay to AI

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Abstract

Teaching AI to play Tabletop Roleplaying Games (TRPG) is a difficult challenge due their negotiable rules and open-ended nature. This is further exacerbated when considering the act of roleplaying, where players do not play or act as themselves, but as a character in a fantasy setting. Previous studies attempting to teach LLMs to play TRPGs do not explicitly discuss role-play in their work, highlighting an absence of a definition in current research. This thesis endeavours in introducing role-playing to AI through developing a prompting method called Control Chain of Thoughts, aimed at teaching it the Dungeons and Dragons alignment system. The prompting method is evaluated through an ablation study where GPT-3.5 is tasked to guess the alignment of characters based on exctracts from D&D gaming sessions. The results indicate a small improvement in GPT's predictions. Further work needs to be done to evaluate if its alignments help LLMs understand roleplaying.

Keywords: *Large Language Models, Tabletop Role-Playing Games, Chain of thought, prompting, AI*

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1 Introduction

Turing (1953) first proposed the challenge of a computer playing and mastering the game of chess. With AI now being able to beat human players in both chess and Go, Ellis and Hendler (2017) proposed to further his challenge by having AI play *Tabletop Roleplaying Games* (TRPG). To highlight the contrast between the two, chess is a turn-based game played by two players with a limited set of mechanics and clear end states. TRPGs, on the other hand, are played in groups, with one player serving as *Game Master* (GM), with rules that are partly negotiated between them, and where the end-goal is not defined. Where the former has a finite set of predictable outcomes, the latter has infinite unpredictable ones.

It might seem curious to mention both Turing (1953) and Goffman (1974) within the context of TRPGs, so their favour for chess will be used as a common ground to clarify their respective importance. Turing wrote of how a computer could play chess against a human and Goffman wrote of what it means for humans to play chess. Both reduced chess into a set of rules, where Turing saw the potential of translating these to be understandable for a computer. Goffman instead used his frame analysis defining a set of rules explaining human activities in real life, such as the playing of chess. Since the rules of TRPGs are negotiated between (human) players, it is then not inconceivable that for a computer to engage in a game of TRPG, it must not only understand the rules of the game, but of the humans playing it too.

Furthering Goffman's (1974) frame analysis, Fine (1983) defines the key aspect of TRPGs to be that of role-playing and describes its function through his different frames, *commonsense reality, game context* and *fantasy*. To engross oneself, one must reach the fantasy frame, which requires limiting the information from the other frames. Fine also identifies two types of players and their playstyles, the role-players, and self-players, where the former is more comfortable with role-playing as their characters. Both types of players sometimes break character, but the self-players do so more often in favour of making their character stronger. When a player breaks character when playing it simultaneously breaks the frame of fantasy not only for said player, but for the others as well.

Large Language Models (LLM) are AIs trained on a large corpus of text-based data to both comprehend and generate text (Ozdemir 2023; Alto 2024). They are therefore highly relevant for TRPGs due to their ability to understand natural language (non-machine-based language) since TRPGs are played through communication between players. GPT (OpenAI 2024) is a decoder LLM that excels in its ability to generate new creative content, an important skill for any TRPG-player, especially GMs.

Current studies looking to tackle the AI and TRPG challenge (e.g. Callison-Burch et al. 2022; Zhu et al. 2023; Liang, Yang and Zhu 2023; Zhou et al. 2023) touch upon aspects of the fantasy frames, but do not empathise them. The most common method used (in this thesis referred to as *Player Action Prediction*) train an LLM on a dataset consisting of TRPG transcripts to then have it predict which character is to act next and what their action will consist of based on prior actions in the given context. Previous studies seem to attempt ways to distinguish both in and out of character actions, but also intentions of players, indicating that attempts are made to distinguish what role-playing is for the AI. However, none seem to use alignments, which are moral characteristics of characters from Dungeons and Dragons (D&D), designed to help guide players to act in accordance to their character, something which a lot of players struggle with. Alignments are closely tied to the frame of fantasy, indicating that they can be a good starting point when defining role-playing to an AI.

This thesis introduces the frame of fantasy using alignment. This is done through developing a prompting method dubbed *Controlled Chain of Thought* (CCoT). Through this, GPT-3.5 is prompted to identify the alignments of D&D characters based on how they act in different scenarios, which are extracted from in-game transcripts. The prompting method is evaluated through an ablation study, where steps of the prompts are removed to discern areas of strength in the method. The results showed an increase in ablation 3's (A3) accuracy by 11 percentage points which was deemed significant when performing a t-test. However, the result was a large decrease in performance from that of the pilot. Upon further qualitative analysis, faults in the method of testing, the AI's reasoning and the testing data used where discovered. Due to this, the prompting method itself cannot be deemed a success, but the results highlight areas of improvement for future work to come.

1.1 Context of Thesis

The research presented in this thesis was conducted during a funded internship at *Cygames Research* (Cygames Inc 2024). All development of prompts and research conducted has been done so using resources and materials at Cygames. This internship is the result of a cooperation between Cygames and the University of Skövde. The author was provided with salary, living accomodations and valuable game industry experience. This research does therefore not only belong to the author, but also to Cygames Research. The research presented in the thesis resulted in a short paper which was accepted at the IEEE Conference on Games 2024 (IEEE 2024), which can be found in appendix I.

2 Background

This section introduces previous studies, subject specific definitions and other reoccurring terms. These will also be available in the appendix as a vocabulary list (appendix J). The first section (subsection 2.1) introduces *Tabletop Roleplaying Games* (TRPG) along with their specific terms such as *Game* Master (GM), *alignments* and the *frames of role-play*. In the second section (subsection 2.2) the TRPG *Dungeons and Dragons* (D&D) is introduced. In section three (subsection 2.3) *Large Language Models* (LLM) are introduced, along with LLM focused terminology such as *prompting*, *zero-shot, few-shot* and *Chain of Thought* (CoT). Lastly, previous studies focusing on the TRPG challenge are presented (subsection 2.4), in this thesis defined as *Content Generation, Game Master Assistance, Narration Challenge* (NC) and *Player Action Prediction* (PAP).

2.1 TRPG

According to Ellis and Hendler (2017), TRPGs function make them unpredictable and consequently make them difficult for an AI to partake in. As an example, they bring up chess, and how its set rules, clear end-conditions and turn-based two-player structure make up a finite set of outcomes. This allows for an AI to compute the best move to take based on the desired outcome of winning. They continue that, for AI to engage in TRPGs they must understand and adapt to their flexible rules along with the social context that influences them. This section focuses on breaking down the structures of TRPGs, as it is needed to understand why they pose as a challenge for AIs. This will mainly be done through adopting definitions and terms from Fine's work (1983).

2.1.1 Rules and Goals of TRPGs

TRPGs are games where players engage in a fantasy setting by roleplaying as different characters they themselves have created. One player, dubbed as the *Game Master* (GM), serves both as a referee, keeping track of the rules of the game (which are often many and complex), as well as a storyteller, guiding the players through a world they have created, often roleplaying as different characters within it (Fine 1983 p. 6). The rules of TRPGs differs between games, but they all mainly use verbal communication where players act as their characters and dice rolls when determine chance-based outcomes and in battles. However, what one can do in the game is not limited to any set rules but negotiated between the players, and the extent of this may vary with some games being more flexible than others. For example, Dungeons and Dragons have a very flexible set-up, allowing a lot of freedom when creating characters and world settings (1983 p.16).

Even though it can be added for engrossment, no game board or figures are used when playing. Instead, most mechanics and game events occur through using dice, pen and paper, and verbal communication. Dice are used to determine success of actions taken by either the players or by the GM. Pen and paper are used to keep track of inventory and stats of the player characters as well as non-playable characters. Most gameplay plays out through players acting out what their characters are doing through verbally. If the GM allows it and it does not interfere with the engrossment of the game, there are no limits to what a player can do (1983 p.185).

TRPGs have a cooperative set-up since they are often played between multiple players, playing "against" the world created by the GM (1983 pp.165). Fine notes that, even though the GM creates the world, and acts as characters whose goals sometimes rivals that of the players, they do not act as an opponent in the way two chess-players would (1983 p.174). As a referee and storyteller, they have authority and overview of the game to create and maintain immersion through making sure rules are followed and that the story is interesting. Fine goes on, explaining that even though some GMs take enjoyment in making the game hard for players (1983 pp.72-73), they will find themselves without them if they are unfair or create boring scenarios (1983 p.107). Simply put, the goal of the GM is to create engaging gameplay for the players, which at times will entail opposing them to challenge them and create suspense.

Unlike most games, TRPGs do not have set end-states such as winning or losing. The over-all goal of TRPGs is that of engrossing oneself in the game through roleplaying. Then there are the smaller goals which are based on how the players engage as their characters in the game. Fine identifies two player types in his study (1983 p.206) self-players and role-players. The self-players are players with the goal of progression through gaining *experience points* (points that is given to players as rewards for achievements), finding rare treasures and growing stronger, even when it means *breaking character* (1983 p.207 and p.211). When a player breaks character they no longer act as their character would and it is quite common albeit undesired as it risks breaking the engrossment of the other players (1983 p. 208). Role-players maintain the goal of engrossment through roleplaying and will act as their characters even if it limits progression (1983 pp.213-214). Note that both player types value the goals of progression and roleplaying, but differ in what they choose to prioritise.

To summarize, TRPGs are role playing games where a GM sets up a scenario and world setting. This along with the general rules of the game determine what players can and cannot do when playing. They play through interacting with the world while roleplaying as characters they themselves have created. Together all are part in evolving and creating the world by manipulating it with their actions and the end goal depends on the storyline created by the and the goals of the players and their characters.

2.1.2 The Frames of Roleplay

Goffman (1974) describes how the dynamic of social interactions are affected by the context, *frame*, which enclose it. He suggests that, depending on the frame, humans will adapt how they act and speak, *keying* themselves (1974 pp. 81-82) in accordance to the frame they are engaging in. Fine builds onto Goffman's' work, stating that not only does frames affect how one acts, but also ones awareness, and exemplifies this through TRPGs through identifying the *frames of roleplay*. They consist of 3 basic frames each encompassing certain knowledge held by the player (1983 p. 194). The first one is that of *common-sense frame*, and is grounded in Goffman's "primary framework" (1974 p. 27). It consists of the understanding and knowledge players have of the real world (1983 p. 186). The second is the *game context frame*, encompassing the knowledge of the game rules and the conventions of the game. Lastly there is the *fantasy world frame*, encompassing knowledge that the character only should have in the fantasy world. Fine (1983 p.186) points out that, in TRPGs the players play as other characters than themselves framing themselves into the frame of fantasy. This means that the knowledge the player uses when playing must match that of their character in the fantasy world's context. In comparison to a game like chess, a player would not adhere from making a move based on the knowledge or background of the pawn they wished to use. The player doesn't consider that a bishop might find it wrong to kill another pawn based on their Christian belief, and whether another pawn is in eyesight of the bishop is not considered either. The knowledge taken in and used is that of the player, common-sense reality, and game context. In TRPGs however, the player must consider the fact that their character is a cleric and might not wish to aimlessly kill other characters, therefor adapting their actions based on their character and their knowledge.

The three frames explain the complexities of roleplaying because, while all players have the knowledge of game rules and of their reality, this knowledge should be inaccessible to the characters they play. This means that players must limit their knowledge to act according to their character (1983 p. 195). What makes this complex is that TRPGs are set up in a way where the players must jump between the frame of fantasy and the frame of game context whenever there is the need to role a dice or query the GM regarding a specific rule (1983 pp. 196-197). Because of this, the player must be able to follow which frame the other players are currently in, so as to not unnecessarily break the frame of fantasy and thereby the engrossment. These frames are important to consider when incorporating AI in TRPG contexts, because depending on the frame, the meaning of information given, and actions taken will differ.

2.1.3 Breaking Character/Frame

When a player *breaks character* they are using knowledge inaccessible to the fantasy frame while still keyed in it. There are times when players need to move from the fantasy frame for the gameplay, such as when making a dice roll, asking about rules or going to the bathroom. These actions are not counted as breaking character, because they are not in keying their action as their character, but as themselves, and thereby actively stepping out of the fantasy frame to engage in one of the other frames. Simply put, the key to breaking character is that of what keying is being used in what frame.

Fine states that, since there are no physical aspects in roleplaying (apart from throwing dice) the physicality and movement of characters is not based on that of the players controlling them, making them easier to play along with, as the characters' physique and movements are clearly separated from the players' (Fine 1983 pp.208 – 211). However, because TRPGs have players act out their characters' actions and statements verbally, their personality can more easily bleed into that of the character (1983 p. 210). Players (especially self-players) also tend to dislike acting in accordance to personality traits and features they do not feel align with their own, such as *moral alignments* and intelligence. As a countermeasure they often play characters similar to themselves (1983 p.196), thereby extending themselves to their character. Because breaking character affects the engrossment of all players it is the GM who decides if they will let a player break character or not (1983 p. 211).

Fine notes a juxtaposition in how TRPGs are set up and their goal of engrossment through roleplay. Since a large amount of the game is based on battling and engaging with characters in the fantasy frame, a lot of the rewards are progression based (1983 pp. 203 - 204). This makes it more rewarding to adapt a progression goal-oriented approach, as it will be more rewarding for the growth of the character. However, it is important to note that, since Fine's work was written in the 80s there have been some updates to some of the TRPGs that were popular at the time. For example, D&D now include inspiration points (Mearls and Crawford 2014), which are points that the GM can give out to players who excel in their roleplay, thereby creating an incentive to remain in character and retain the engrossment.

2.2 Dungeons and Dragons

The first edition of *Dungeons and Dragons* (D&D) was published by Arneson and Gygax 1974 (Tresca 2010, p. 62). The game, being based in a medieval fantasy setting, featured four races (humans, dwarves, elves, and hobbits) and three classes (fighting, mages, and clerics). Using these, players could create their own characters to roleplay with when playing. D&D is famous for allowing for a lot of flexibility for players and GMs, since there is no designed social structure in the game (Fine 1983, p.16). Instead, players and GM have free access to create scenarios and worlds based strictly on their preferences. Over the years the game has evolved, now being on its fifth edition (Mearls and Crawford 2014) with 9 playable races (in the original version) and 14 playable classes. While still maintaining a medieval fantasy setting, it now has more roles to play, and lots of expansions for the GM to use.

When playing D&D, players are first tasked to create their character. Here they choose what race, class, and background their character shall have, making it possible for them to have it match their preferred gaming style. They are also tasked to choose an alignment (the moral and social attitude of the character) and are often encouraged to tie this into the background chosen. The player, by rolling dice, determines the value of their character's stats: strength, wisdom, dexterity, intelligence, constitution, and charisma. Based on class and race the player will also gain extra points to their stats affected by certain inherent traits said race or class has. These stats are then used in gameplay as *ability scor*es and can help players depending on the situation. When a player has a high ability score, they will be able to add points to their dice rolls, whereas if their ability score is low, they must retract points. An example for a situation where the ability scores can help is for example when a player needs to open a heavy door. If their character's stats in strength is high their ability score will be higher too. This enables them to succeed even with a lower value dice roll since the points of the ability score is added onto said value.

2.2.1 Alignments

The alignment system of D&D as it is now has existed since the first edition (Arneson and Gygax 1974). Alignments consists of two variables, one defining a character's attitude toward societal norms and authority *lawful-chaos* (lawful, neutral, or chaotic), and the other defining their moral values *good - evil* (good, neutral, or evil). These together form nine different alignments (figure 1). The alignments serve as keying tools for players when roleplaying and are normally static, meaning that they do not change throughout the gameplay.

In *D&D Basic Rules* (Crawford, Mearls, Perkins 2018) the characteristics of lawful-chaos are defined as a characters' "attitude toward society and order". Someone who is lawful value the order of society, such as rules, laws, and social norms, whereas someone who's chaotic mainly value their own opinion and morals and do not transcribe to social rules. This does not mean that characters who are lawful always have good morals and chaotic characters have bad, it merely indicates their stance and view on society. A character can also be neutral, meaning that they value the order and rules of society as best they can, but also believe that societal rules might not always be absolute truth.

Good-evil defines a character's morals. A character who is good will do what they deem is the best and nicest thing to do and someone who is evil will do the opposite. An important note is that what they base their morals on is influenced by their attitude toward society (law-chaos). For example, a character who is chaotic good might find it okay to steal if it will help someone else, whereas someone who is lawful good will not approve of it at all. If a characters morals are neutral they have no stronger feelings about what is right and wrong, and tend to avoid making moral decisions if possible.

Lawful good (LG) creatures can be counted on to do the right thing as expected by society. Gold dragons, paladins, and most dwarves are lawful good.

Neutral good (NG) folk do the best they can to help others according to their needs. Many celestials, some cloud giants, and most gnomes are neutral good.

Chaotic good (CG) creatures act as their conscience directs, with little regard for what others expect. Copper dragons, many elves, and unicorns are chaotic good.

Lawful neutral (LN) individuals act in accordance with law, tradition, or personal codes. Many monks and some wizards are lawful neutral.

Neutral (N) is the alignment of those who prefer to steer clear of moral questions and don't take sides, doing what seems best at the time. Lizardfolk, most druids, and many humans are neutral.

Chaotic neutral (CN) creatures follow their whims, holding their personal freedom above all else. Many barbarians and rogues, and some bards, are chaotic neutral.

Lawful evil (LE) creatures methodically take what they want, within the limits of a code of tradition, loyalty, or order. Devils, blue dragons, and hobgoblins are lawful evil.

Neutral evil (NE) is the alignment of those who do whatever they can get away with, without compassion or qualms. Many drow, some cloud giants, and yugoloths are neutral evil.

Chaotic evil (CE) creatures act with arbitrary violence, spurred by their greed, hatred, or bloodlust. Demons, red dragons, and orcs are chaotic evil.

Figure 1 The Alignment Description from Dungeons and Dragons Basic Rules (Crawford, Mearls, Perkins 2018, p.35 – 36)

2.3 Large Language Models

Large Language Models (LLM) are AI models trained on a large corpus of text data and is a subfield in neural networks (Ozdemir 2023). They store their training data in something called *parameters*. Parameters store the pre-training data in the form of *tokens* (Ozdemir 2023; Hu et al. 2021). A token is a unit for the smallest piece of information of semantic meaning and are used as inputs when predicting the next word to output (Ozdemir 2023). They can represent a character, a word, or even sub-words, which is common in cases where an LLM does not recognize parts of a word (Alto 2024). As an example of a word being broken down and tokenized two sub-words take the word nyancat, where the LLM does not recognize nyan, but does recognize cat. It separates the two, creating two tokens nyan- and -cat, marking that they are connected and making them into sub-words, where one is a word that it recognizes, which gives the LLM some understanding of what the whole word means. The more data that is stored the more parameters are needed. Larger LLMs often have over billions of parameters where training data is kept.

LLMs use neural networks to predict the next sequence of tokens to output based on data it has trained on. When a user inputs a request by writing text the LLM creates an output based on said input, using a statistical prediction based on the data it has trained. However, the output might not always match what the input asked for. In these cases, there are ways to prompt and train the LLM to create a more accurate output. This section will go through the architecture of LLMs, what GPT-3.5 (OpenAI 2024) is, and what *finetuning* and *prompting* entails.

2.3.1 The Transformer

2017, Vaswani et al. released their paper *Attention is All you Need*, where they showcased a new model architecture for neural network-based AI called the *transformer* (Vaswani et al. 2017). It was this architecture (figure 2) that revolutionised AI development and is now the basis for the architecture for all LLM.

The transformer consists of an *encoder* and a *decoder* as well as the *attention function* (Vaswani et al. 2017). The attention function looks at the position of the words in an input sentence, looking for semantic or contextual dependencies. The encoder takes in and encodes the words as tokens, giving each a vector position based on their semantic and contextual meaning. The distance between where these encoded words are stored in the vectors is measured using the cosine of the angle between them (also known as their *cosine similarity*) (Alto 2024). If two tokens are angled pointing in the same or a similar direction they have a similar meaning, and the opposite is true if they are not. The decoder predicts the next word when generating a text output. It does so through training on a large corpus of text, while also taking in the relationship of the words made in the input through the attention function (Vaswani et al. 2017).

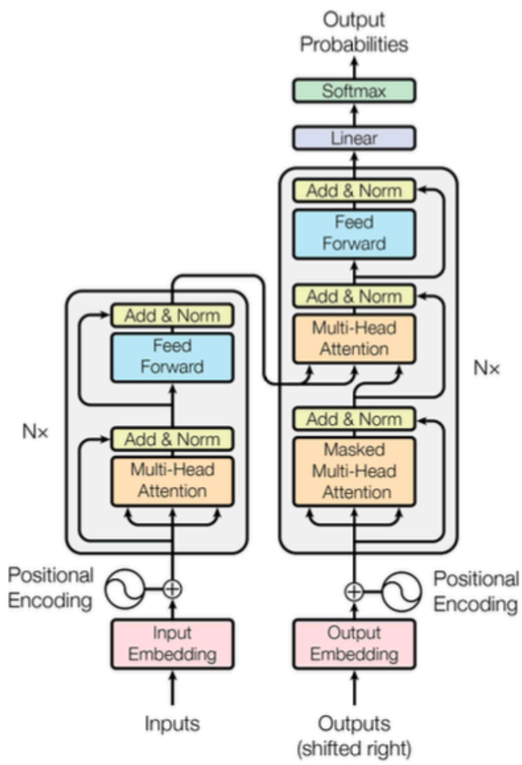

Figure 2 An illustration of the transformer architecture taken from Vaswani et al. (2017)

This means that the transformer, using the encoder, decoder and the attention function takes in an input, for example a sentence in French, and then encode each word into different vectors and layers based on their usage in the sentence, then decode these in English, generating a translation (Vaswani et al. 2017). However, nowadays a lot of LLMs only use either the encoder part of the architecture or the decoder, with some exceptions using both (Alto 2024; Ozdemir 2023) such as T5 (Google 2024). Encoder based models, such as BERT (Google 2024), are great for classification tasks because of their semantic and contextual understanding (Ozdemir 2023), whereas decoder models, such as GPT-3.5 (OpenAI 2024) are great for generating free-text based on a given context (Ozdemir 2023). In this thesis the decoder model GPT-3.5 (OpenAI 2024) is used.

2.3.2 GPT and OpenAI

GPT (OpenAI 2024) is an LLM developed by OpenAI. The name stands for Generative Pre-Trained Transformers (Alto 2024), referring to its decoder-based architecture. Over the years, OpenAI have created several different models, their newest one as of writing being GPT-4o (OpenAI 2023). All OpenAI's models are pre-trained on large amount of text data, with billions of parameters. They are perhaps most famous for *ChatGPT* (OpenAI 2024), a GPT-based chatbot that users interact with through inputting prompts, often in the forms of questions, which it then answers. Because of its large training data, it can help users with coding issues, planning their days, what they should feed their dog, and so on (Alto 2024). These close interactions with users enable GPT to also train through reinforcement learning from human feedback (Open AI 2023).

OpenAI has a developer platform referred to as API (OpenAI 2024). Here they share different types of documentation for developers to use different prompting methods, their embeddings as well as how to finetune a GPT model (Open AI 2024d). Here they also share the playground (OpenAI API 2024e), where users and developers can test out different prompting methods, without having the LLM train on the inputs. However, at the time of writing this thesis, it is not yet available to use GPT-4 in the playground or for finetuning.

2.3.3 Finetuning

Because LLMs are trained on a large amount of text data, they are very general, making them suitable for most tasks. However, if one wish to use and LLM for a specific task this generalness is not enough and it needs to be finetuned for the intended task. When finetuning an LLM, you change its behaviour through altering its parameters by providing it with your own dataset (Ozdemir 2023; Alto 2024). This can be helpful to make the LLM more precise, however depending on the model used fine-tuning can be a very resource and time-consuming task, since all parameters must be updated. For a model like GPT-3.5 this would entail updating over billions of parameters (OpenAI 2024e), which would require a lot of data.

There are ways to bypass having to alter all parameters and that is through adding new parameters (Alto 2024: Ozdemir 2023; Hu et al. 2021). This will not require the same size in dataset while still retaining the original pre-trained data of the model. An example of this is the *low-rank-adaptation model for LLMs* (LoRA) created by Hu et al. (2021). This both reduces the amount of data required to fine-tune it, since it allows for adapting how many parameters to include and bypasses the need to have access to the source code of a model. Using a method like LoRA enables adding additional data, further specialising a pre-trained LLM with this, without having to change any of the existing parameters.

OpenAI (2024d) allows for these kinds of finetuning, but only on certain models of theirs, and they also recommend users to first testing out their data through prompting (OpenAI 2024d). This way, one can first determine if the dataset to be added to the model is enough by prompting. If not the prompting will highlight areas of weaknesses that can be addressed before finetuning.

2.3.4 Prompt Engineering

Prompts are user created inputs explaining a desired output from a pre-trained LLM (Phokela et al. 2023). The development and structuring of these prompts is called *prompt engineering* and depending on the structure of a prompt the quality in outputs may differ and often the same output does not generate the same answers across different LLMs (Ozdemir 2023). There is already a lot of different prompting methods and some of the most reoccurring ones in previous studies are *zero-shot* prompting, *few-shot* prompting, and *Controlled Chain of Thought* (CoT) prompting.

Zero-shot prompting, few-shot prompting are similar approaches, where the user prompts an LLM by stating what kind of output they want (Brown et al. 2020). In zero-shot prompting, the user gives no example prompts, but simply inputs the task desired for the LLM to do. An example of zero-shot would be a user asking what dog breed they should get based on their personality. In few-shot prompting the LLM is prompted with one or more input and output examples, to demonstrate the task asked of the LLM (Wei et al. 2022). Because decoder LLMs generate text through predicting the next word most likely to follow in a sentence (section 2.3), few-shot helps minimise the possible outcomes by giving examples of the desired task. To expand on the prior example, a user using few-shot also includes examples of their friends who have dogs, describing their personalities and stating what dogs breeds they have before asking what breed would be suitable for them. In cases where only one example is used it is also commonly referred to as a one-shot prompt (Kojima et al 2022). For the sake of simplicity, this thesis includes one-shot prompts in its few-shot prompt definition.

Few-shot prompting is a method that can be used in several different ways and is quite flexible since it includes examples of the desired output from the user. For example, Zhou et al. (2023) uses few-shot prompting as the base of their Theory-of-Mind approach. In their study they try to look at the intention of the GM through prompting an LLM with an excerpt from a D&D session, including the GM's guiding utterance and the following ability check of the player. After this it is given another utterance made by the GM, made to guide a player and then tasked to predict what the intentions of said utterance was (figure 3).

"*The following is a conversation that happened in a game of Dungeons and Dragons: [Context] [DM Text] [Player Name]:[Player Ability Check] Question: What do you think that the DM intentds to do by mentioning [Extracted Guiding Sentence]?*

Figure 3 An example of the Theory of Mind prompt from Zhou et al. (2023)

Brown et al. (2020) mentions that the main advantage of few-shot learning is that it creates a major reduction in the need for task-specific data, since the user provides the LLM with the data through their prompts. However, some disadvantages are that prompting requires the user to input a lot of tokens, making it costly, and it still requires some task specific data, albeit less than when finetuning a model.

Another prompting method is chain-of-thought prompting (CoT), and it has its basis in arithmetic problem solving and is a way of prompting where the LLM is prompted to break a task down into several smaller ones (Wei et al. 2022). Because decoders use next word prediction to generate text, breaking down larger tasks allows for additional computation, and further specificity when predicting the next word. Since the LLM is outputting its *reasoning* chain (chain of thoughts) the user can also see faults in the *reasoning* thus indicating areas of improvements in the prompts. Wei et al. (2022) also found that CoT prompting had great effect on common-sense *reasoning*, which is an ability that is crucial when understanding human conversation. Note that Wei et al. (2022) uses the term *reasoning,* something which will also be employed in this thesis, however, note that this does not mean the same type of reasoning humans do. In this case, reasoning refers to the break-down of an LLMs calculations in predicting the next word to generate, and not human reasoning, as the latter is something not (yet) possible for AI.

There are different types of CoT prompting depending on if they use a zero-shot, or few-shot method. In the example of Wei et al. (2023) they present their LLM with a reasoning example, using a few-shot CoT prompting method. First, they give an arithmetic question, then they add an example in the form of an answer including the thought process of how the answer was reached (figure 4). This way the AI is taught how to reason itself to the correct answer through the prompt.

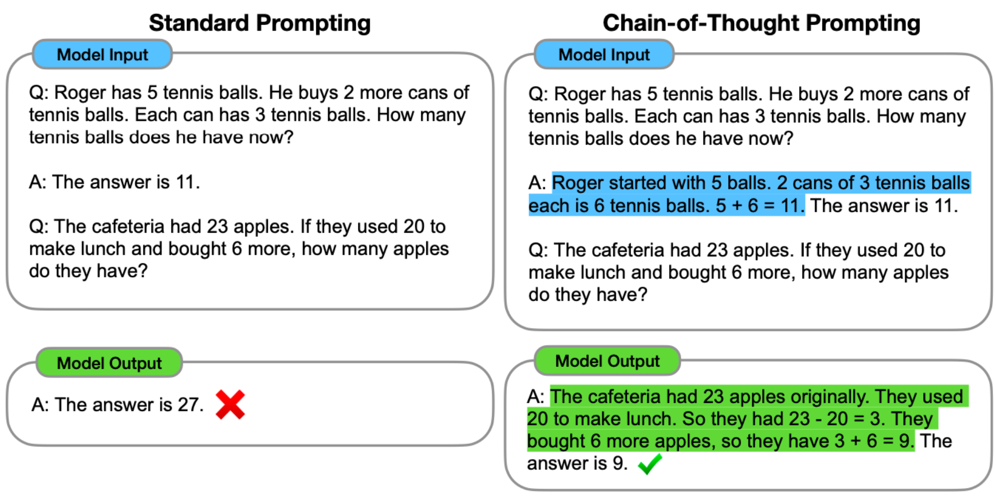

Figure 4 Example from Wei et al. (2022) showing the differences in the output generated through a one-shot prompt and a one-shot CoT prompt.

It is also possible to use CoT with a zero-prompt method where no reasoning examples are given, and instead asking the model to "*think step-by-step"*(Kojima et al. 2022; Wang et al 2023a). This way, the LLM creates their own reasoning chain, which is less labour intense since it does not require example data, however, it also risks the AI drawing the wrong conclusion during its reasoning chain. In the case of Kojima et al. (2022), they first had an LLM answer questions using CoT, and then used those instances leading to the correct answer as training examples in prompts. This way, they first started with a zero-shot CoT approach to generate reasoning examples, and then used these for their few-shot CoT prompts.

Because of how CoT prompting breaks down the task for an AI it can take on bigger tasks. For example, Wang et al. (2023b; 2023c) was able to create a generative agent that could play *Minecraft* through using an LLM as an interactive planner to break down big goals to smaller ones using zero-shot CoT. For example, if the agent was to build a bed it had the task broken down to collecting wood, wool and creating a workbench (figure 5). Whenever it encountered an obstacle, it was prompted with a new zero-shot CoT prompt to generate a new set of subgoals to have the agent adapt to obstacles occurring when playing (e.g. encountering enemies).

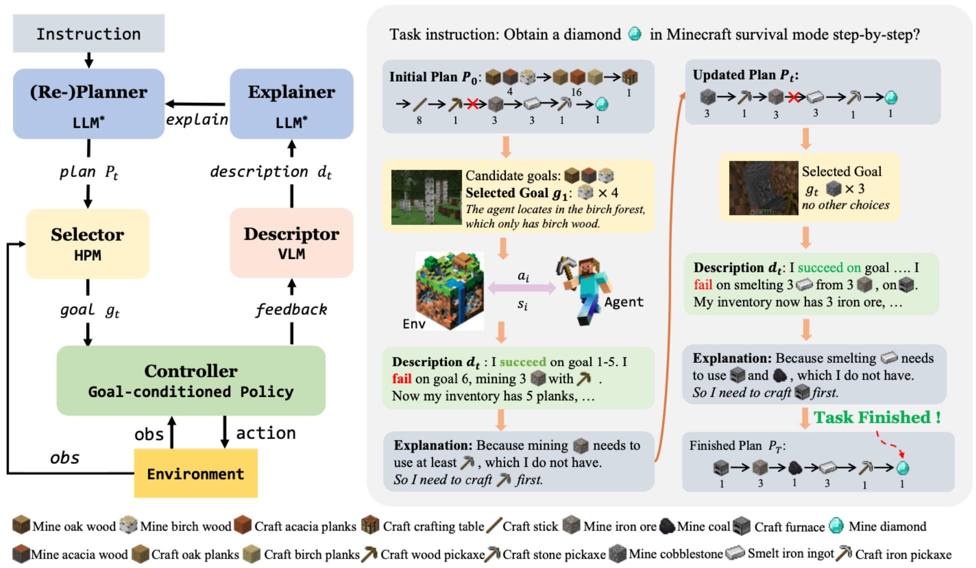

Figure 2: **Overview of our proposed interactive planner architecture.**

Figure 5 Example of the architecture of the LLM-based generative agent from Wang et al. (2023b)

CoT has also been used in previous studies to elicit so called "self-reflection" among LLM (Wei et al 2022; Park et al. 2023) This is done by giving the model prompts formed as questions regarding recent information it ascertained. Through this it is prompted to create a higher-level abstract thought to generate a reflection summary of the information. For example, in the study by Park et al. (2022) they give the example where an LLM is prompted with information about Mr. Mueller and then asked to infer high-level insights based on this information and then summarized said insight into a reflective thought (figure 6).

```
Statements about Klaus Mueller
1. Klaus Mueller is writing a research paper
2. Klaus Mueller enjoys reading a book
on gentrification
3. Klaus Mueller is conversing with Ayesha Khan
about exercising [...]
What 5 high-level insights can you infer from
the above statements? (example format: insight
(because of 1, 5, 3))
```

Figure 6 Example of a zero-shot CoT prompting aimed to elicit self-reflection from the LLM (Park et al. 2023)

Note that there are several more different types of prompting methods developed and being developed currently (Sahoo et al. 2024) (figure 7). However, at the time of writing this thesis, many methods have yet to be peer-reviewed and/or tested through replication studies. Zero-shot, few-shot and CoT, are peer-reviewed prompting methods that have existed for a longer period. These prompting methods are often the basis of the new methods that have been or are being developed, proving their success in the prompt engineering field, making them relevant to this thesis.

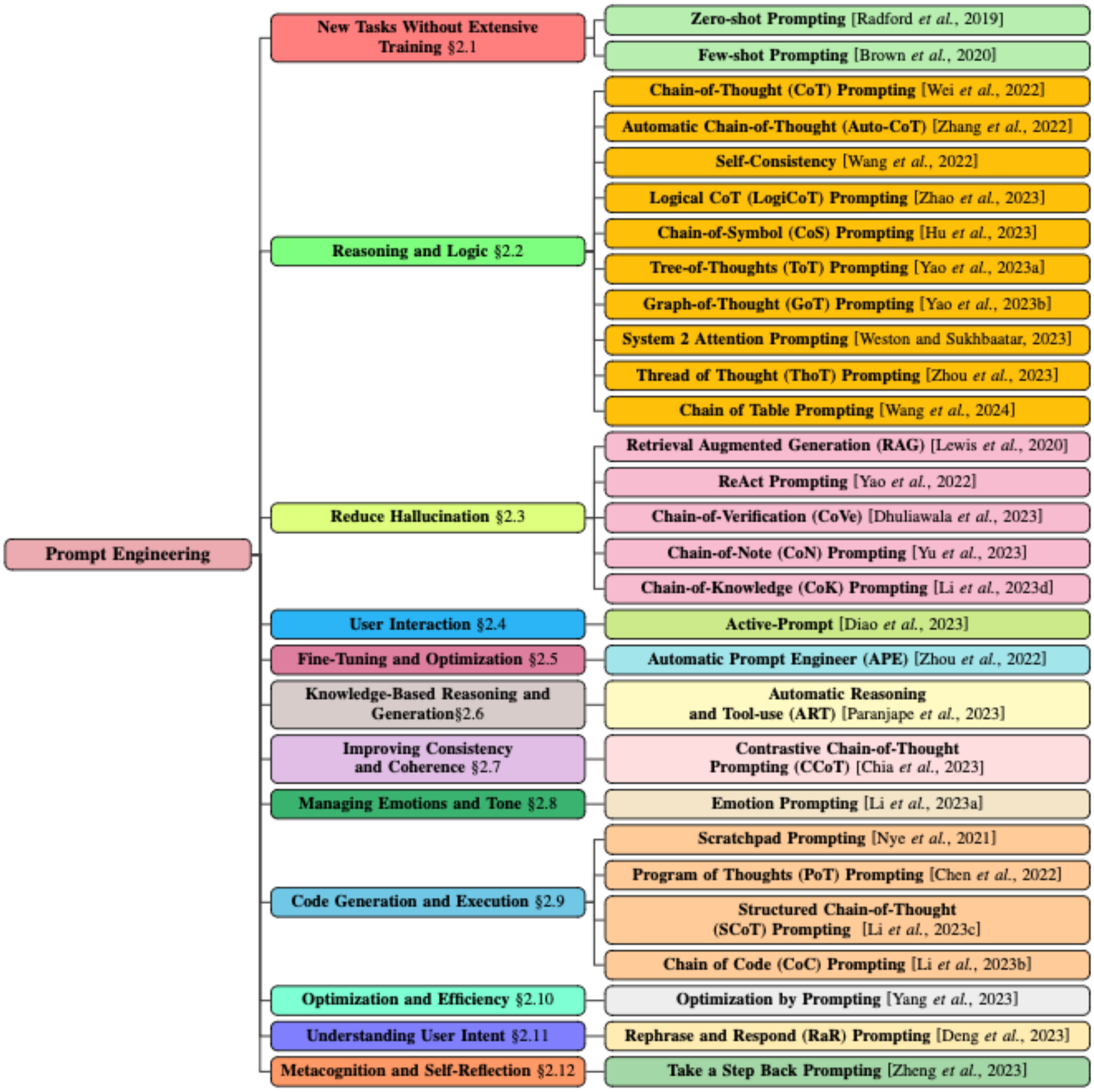

Figure 7 A taxonomy of the different types of prompting methods that have been/are being developed (Sahoo et al. 2024)

Prompting engineering is a great method when wishing to use the data existing in a pre-trained LLM while still adapting its behaviour. It can be seen as a good first stepping stone when wishing to finetune an LLM, since it requires less data. However, using prompt engineering also has its limits. Firstly, when designing prompts, one must be consistent in their design to not generate inaccurate outputs (Phokela et al. 2023). Secondly, few-shot prompting methods requires lots of extra tokens when giving the model examples, making it less cost effective. The more precise one wishes to be the more data must be shared with the model, amounting to more tokens, thereby making less suitable for larger tasks.

2.4 Previous studies conducted on LLM and TRPG

With Ellis and Hendler's challenge (2017) and the increased interest in LLMs, more research has started to emerge, looking at how said challenge can be tackled. Previous studies on the area present different methodologies and strategies, some of which will be presented in this section. These studies have been defined as *Content Generation*, *GM Assistance*, *Narrative Challenge* (NC) and *Player Action Prediction* (PAP) studies. Content Generation and GM Assistance have similar approaches, where the LLM's ability of generating and re-contextualising creative content is used. The differences between these are that Content Generation studies have a static approach, generating content from outside the game context through training an LLM on TRPG rule data. GM Assistance studies use a dynamic approach where content generation is used to summarise game states and make new decisions during gameplay. Their methods and evaluations are therefore different, but both focus on the same capabilities that exist within decoder-based AIs.

The most common study type is that of the PaP studies, where they create a dataset based on TRPG game-session transcripts to train a model to predict which player is to act next and what their action will consist of (Callison Burch et al. 2022; Zhu et al. 2023; Zhou et al. 2023; Liang, Yang and Zhu 2023). This method derives from the Narrative Challenge studies (Louis and Sutton 2018; Rameshkumar and Bailey 2020; Si, Ammanabrolu and Riedl 2021), where TRPGs were used as training datasets in different ways to test the narrative understanding of Language Models, looking at specifically character dynamics. The distinction between Narrative Challenge and PAP studies is that the former is using TRPG transcripts as an attempt to test the narration ability of an LLM, and the latter to test its ability to understand and partake in TRPGs.

2.4.1 Content Generation

One of the strengths of LLMs are their ability to generate text-based content based on user input (Zhu et al. 2023b). They can also be finetuned based on datasets to generate more accurate results (Wang et al. 2023a). When generating content for TRPGs, there are often lots of rules and mechanics to adhere to, making it a time-consuming job for players to do themselves (Ma et al. 2023; Zhu et al. 2023b; Góngora et al. 2024). There can also be need for new content during a game session when players act unexpectantly, requiring fast thinking, which can be difficult for a human, but effortless for an LLM.

Newman and Liu (2022) use pre-existing spells from the D&D rulebook, as well as player-made ones posted on a D&D forum (dandwiki.com) as their dataset for training an LLMs to generate new rule-adhering spells. They test 5 of their LLM-generated along with 5 player-made spells by having 14 experienced D&D players read them. Afterwards they have to guess *what* they thought had made the spells, how well they conform to the rules of the game, if they would allow it when playing and if they have seen it before. While at times being able to tell when an AI had made a spell, they still approved of their mechanics and could see themselves using them while playing. At two occurrences, players marks AI-made spells as ones they recognised from before, even though they did not exist prior to the experiment, which Newman and Liu (2022) claims show that LLMs can achieve a semantic likeness to the core material of D&D. It is however important to note that given how decoder LLMs generate outputs (subsection 2.3), it is possible that this could in fact be an example of the LLM only rephrasing an existing spell due to lack of information. It is difficult to determine this however, as the dataset or the list of spells are not shared in their study.

There are examples of studies focusing on content generation using LLMs in game contexts, which do not focus on TRPGs, but whose methods could be useful in a TRPG context. Examples of such are Volden, Grbic and Burelli (2023) who have an LLM generate rules for a narrative puzzle game for children, Al-Nassar et al. (2023) which generate quests with connected storyline and Wang and Gordon (2023) who train an LLM to partake in a narrative game where it is prompted to generate a short story based on specific requirements. Despite not focusing on TRPGs, these studies are good examples of different ways to generate new game content using an LLM.

Common issues when looking at content generation made by AI, is the issue of LLMs generating false content that does not exist in the given context, also referred to as *hallucinations* (Zhu et al. 2023b; Zhou et al. 2023). There is also occurrences of unsuitable explicit depictions and/or language (Al-Nassar et al. 2023; Zhu et al. 2023b). These issues make it so that content generation in AI is still at a level where human intervention is needed.

2.4.2 GM Assistance

The role of the GM is an information dense one, and this is where AIs can be helpful. Because of AI's ability to take in vast amounts of information as input and create an output based on this, there is great use of them as GM Assistants (Zhu et al. 2023b; Gongora et al. 2023; Santiago et al. 2023). Zhu et al. (2023b) creates a GM assistant called CALYPSO, which can help generate new ideas, retrieving and summarizing information, create new content and aid the GM in making game decisions. To design the interface for the assistant they arrange a workshop with 7 experienced D&D GMs to share what they look for in an assistant. Based on the results of the workshop the assistant is given an *Encounter Understanding* mechanic, *Focused Brainstorming* mechanic, and an *Open-Domain* baseline. *Encounter Understanding* includes a summarization mechanic, enabling GMs to summarise lore and information based on the D&D rulebook by pressing a button. They also include a canon of medieval fantasy when training CALYPSO, since some information in the rulebook expects the GM to have general knowledge of the fantasy genre beforehand. The *Focus Brainstorming* mechanic allows for GMs to input questions about encounters to receive strategic advice or descriptions of what the battlefield looks like. Lastly, the *Open-Domain Baseline* consists of *Chat-GPT-3* posing as a guide for the players and GM to ask general questions about rules, alleviating the GM's work as a referee. Because it is part of the CALYPSO interface it has access to all the game session data, enabling it to answer according to the players' scenarios.

When evaluating the assistant, they have 71 experienced D&D players divided into 21 groups of players, with one GM in each, play a session of D&D over discord. This way, all communication will be textual, enabling using it as input for the assistant. The setting of the sessions is pre-defined by the authors to avoid diversion when testing. The 21 GMs participating is then interviewed. Some areas of strengths are CALYPSO's ability to help GMs understand the current context in the game through its summarisation tool. They also find it good for distilling long information, and high-level descriptions aid in decision making, and in creating understanding of enemy motives and strategies. Areas of weaknesses are that, despite distilling information correctly most times, it still only rephrases text from the rulebook rather than summarising it. CALYPSO also performs worse when generating information not included in the D&D rulebook, needing thorough prompting to do so. When using mechanics not incorporated in CALYPSO, such as dice rolls it does at times receive error messages or hallucinate. Despite its weaknesses CALYPSO is appreciated by GMs, although Zhu et al. (2023b) does make the point that, despite working as a helpful tool, they find that AIs are not ready to be used as a stand-alone GM.

2.4.3 TRPGs as a Narrative Challenge

Some studies include aspects of TRPGs in their research, but without the purpose of training an LLM in tasks focused on partaking or assisting in playing TRPGs. Instead, these focus on different narrative challenges such as continuing a story through generating dialogue or summarising information, excluding irrelevant content and noise when doing so. In this thesis these are dubbed Narrative Challenges, and their methods are often referenced in that of PAP studies as the major inspiration. This section will introduce these studies briefly to give context to the method present in the PAP studies, as well as examples of how TRPG transcripts can be used as datasets.

The first Narrative Challenge study is that of Louis and Sutton (2018). They argue that previous studies, testing AI's narrative understanding lack focus on character dynamics. They propose TRPG transcripts as good datasets for training language models, as TRPGs create narratives through the player-characters' interactions with each other. The transcripts are also naturally structured to have all utterances made paired with the character who made them, data which would require excessive cleaning if collected from conventional novels. To test their theory, they do an *ablation study* where they test their own *language model,* LM (like LLM but with smaller dataset), which has been trained on TRPG transcripts attained from an online roleplaying forum to predict the next utterance to come. In an ablation study, parts of an AI's structure or prompts is removed or altered, to evaluate their effects on performance. For their model, Louis and Sutton (2018) observe how character information and prior sequence of actions affect the *perplexity* score. Perplexity refers to the range of possible predictions for an LM when generating an output, where lower perplexity means a lower range and higher precision. They found that including both which character is to act next and prior actions improved the perplexity score.

Inspired by Louis and Sutton (2018), Rameshkumar and Bailey (2020) create a dataset with the purpose to train LLMs to summarise live text-based conversations, while excluding irrelevant noise. As data they use transcriptions of *Critical Role* (Critical Role 2024), a livestreamed show where actors play D&D together. They believe that transcriptions of live play-session simulate reality more, as they contain out of context conversation, background noise and spontaneity. They then use summarisations taken Critical Role's fan-page (Critical Role Wiki 2024) pairing them with the relevant transcripts as examples of summarisations, thus creating the Critical Role Dataset.

Si, Ammanabrolu and Riedl (2021) use the Critical Role dataset (Rameshkumar and Bailey 2020) to modify the method of Louis and Sutton (2018). They use the dataset to train their bi-encoder LM (figure 8) to take in two types of data: context (current conversation) and candidate utterance (possible response) to predict which character is to act next and what they will say. They evaluate the model's performance through an ablation study, where the extent of context given was altered, ranging from one to ten chunks of text. To also test how relationship information affected the performance, they do two versions of the ablation study, one with relationship information and one without. Results shows that the model has an increase in performance with longer historical context. However, this increase is at its largest between using one and two chunks of texts, to then plateau as more is added. Including relationship information has a similar effect on performance to including character backgrounds.

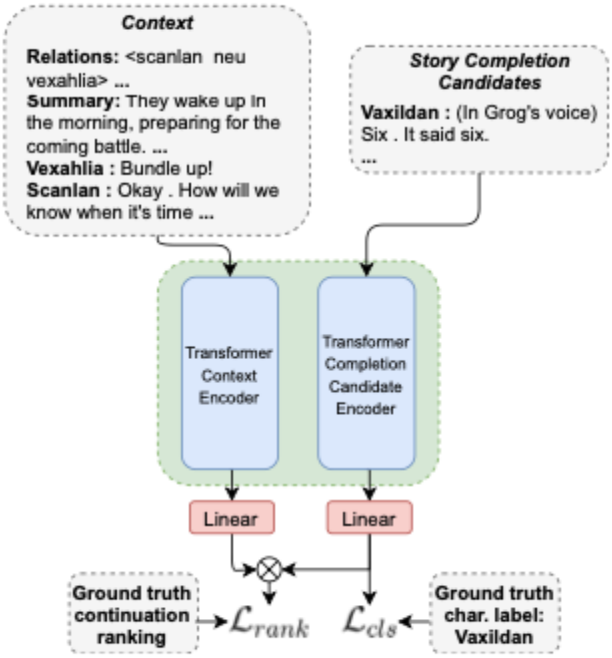

Figure 8 The architecture of the bi-encoder by Si Ammanabrolu and Riedl (2021). It first encodes data as vectors to then pair the context with the best candidate utterance based on their comparing their cosine similarity.

2.4.4 Player Action Prediction Studies

PAP studies focus on training LLMs to play TRPGs by creating training-datasets from TRPG transcripts. These studies (Callison-Burch et al. 2022; Zhu et al. 2023a; Zhou et al. 2023) often reference Narrative Challenge studies (subsection 2.4.3) as the starting point of seeing LLMs as the natural solution to the TRPG challenge posed by Ellis and Hendler (2017). PAP studies therefor adopt some methodology from the Narrative Challenge studies, such as evaluating how the dataset improves the LLM's ability to predict the next player to act and what it will do, like Si, Ammanabrolu and Riedl (2021). One difference is that, since PAP studies aim to have LLM interact with humans when playing TRPGs, they often include subjective evaluations with human testers.

Callison-Burch et al. (2022) create a dataset based on a play-by-post forum called *D&D beyond* (D&D Beyond 2024). To clean up the data to only consist of relevant information, they create game-states based on heuristics made by themselves. The states are defined as both dynamic and static, where dynamic represent changeable states, such as health points or being in and out of battle and where static states remain the same throughout the game, such as names, race, and pronouns. The heuristics defined the conditions for when a state is to be labelled. For example, names of characters are defined as the name most often referred to, the GM is assumed to be the first player speaking and combat scenarios are labelled through looking for initiative rolls, and attack rolls that are followed by initiative rolls. This way they can automate the cleaning of data. To define when a player is speaking in and out of character, they collect transcript data from another play-by-post forum, where chats consists of two versions: one with in character communication and one with out of character communication. They then use these transcripts to train a classifier to label their D&D Beyond dataset as in- or out of character actions.

For their testing they use a decoder based LLM trained on conversations. When evaluating they use an ablation study with five different model versions, where the extent of the iplementation of game-states vary. For their evaluation they use both subjective and objective testing. For their objective testing they look at the datasets effect on perplexity score as well as the token accuracy score. Those ablations using game states, whether it be for only the current turn or all turns, performs, where the former performs slightly better on both scores. For the subjective evaluations they look at how the states and amount of context affect the outputs sense-making, specificity and interest. To do this they use 6 raters, 5 of which have played D&D before (the sixth had not) rate 500 predictions made by the model. They rate them through answering three questions: “does the response make sense” (yes/no), “is the response specific” (yes/no) and “how interesting is the response” (10-point Likert scale) (Callison-Burch et al. 2022 p. 6). They all rate 500 system outputs. Results show that, apart from the reference response, those ablations using game states performed best on interest, where including previous and current context had similar effect. However, the ablation only using the dataset performed best on specificity and sensemaking. It is important to note that the scoring is quite similar for all models, with the baseline performing the worst, but still with similar score to that of the others. Callison-Burch et al. (2022) highlight limitations regarding the heuristics of the game-states as they were made by them. However, they believe that the models’ ability to track game states can be a helpful tool for first time players and GMs and propose that, since their method of including game-states improves the LLM’s outputs, it thereby also improves its understanding of the game session.

Building onto Callison-Burch et al. (2022), Zhu et al. (2023a) using the same method, but base their game-state on the heuristics of D&D’s *character sheets* (sheet of paper where players document information about their characters). They also collect transcripts from Avrae, a chatbot that assists players to play D&D online on *Discord* (Discord 2024). Avrae contains all information about characters, tracks combat states and has the function of rolling dice for players, automatically adding their ability scores (subsection 2.2). This way Avrae can generate a command based on player input. For example, if a player wishes to attach an enemy, Avrae will output what weapon is used, the dice roll and the added ability score. Zhu et al. (2023a) use the heuristics to divide the Avrae transcripts into different game-states. They then reuse the in- and out of character chat data from Callison-Burch et al. (2022) to distinguish between in- and out of character actions. Zhu et al. (2023a) divide their action prediction into two tasks, first one being *Utterance to Command*. The Utterance to Command action outputs an utterance to Avrae to see if it generates the desired command. The second one is *State to Narration* task, where the model predicts the next action made by a player given the current context. Both tasks are evaluated through ablation studies.

For the Utterance to Command task, they do four different ablations where the amount of examples from the dataset given and the inclusion of game states vary. During evaluation they evaluate how the many of the outputs (n=1000) are accepted by Avrae and how many lead to the desired outcome. They also include a qualitative analysis where the LLM is prompted to generate an action based on a spell to see what information the LLM considered when doing so. The ablation including examples from the dataset but no game states performs best, accurately keeping track of the states of the characters. The qualitative analysis show that, using said ablation, the LLM considers characters' hit points when casting, aiming for those with lower, indicating a form of strategizing. For the State to Narration task the model is prompted to summarise the states of the game in a narrative manner. They use four different ablations where the extent of character and context information varies. For objective evaluation, they look at the perplexity score and semantic similarit. In the subjective evaluation they have 45 experienced D&D players evaluate the outputs and dataset references using the same questions presented in Callison-Burch et al. (2022). The ablation using most information performed best in sense making and perplexity, whereas the ablation only including the last 5 turns and descriptions of actions taken scored best on semantic similarity. When looking at the subjective evaluation, both of these ablations outperform the other models, even the dataset reference. Zhu et al. (2023a) suggest that the reason for the dataset reference scoring lower than the models could be due to them often referencing background knowledge not included in the context of the extract that the testers had access to.

Zhou et al (2023) suggests that one of the key roles of a GM is to guide the players throughout the game. Building onto the methods used by previous studies (Si, Ammanabrolu and Riedl 2021; Callison-Burch et al. 2022; Zhu et al 2023a), they propose a dataset which includes intentions of GMs. They do this through introducing a new task they dub as *G4C: Generating Guidance in Goal Driven and Grounded Communication*. Using the dataset of Callison-Burch et al. (2022), Zhou et al (2023) extract examples of GMs giving guidance to players. First, they have humans go through dialogues from 5% of the dataset and label which GM utterances contain guidance and to then imagine what action they would take based on said guidance (how many of these was created is not mentioned). They then use a *Inverse Dynamics Model* (IDM), a model trained on predicting what action led to a certain outcome based on the previous state of the system. The IDM is then trained to identify and extract key-guiding sentences from the data. Like the classifier in Callison-Burch et al. (2022), they have the IDM label the guiding utterances in the dataset to automatically extract them later.

To model the intents of the GM, Zhou et al. (2023) create a Theory of Mind prompting method, where they prompt GPT-3 to explain the intent of the GM given the current context. In the prompt they include the context (the past turn), the GM utterance, the player's name, and corresponding action (defined as ability check). This is followed by a question regarding the intents of the GM's utterance. Using these intents as data, they then create an intention-generating model that the virtual GM model use as inputs to generate guidance outputs. To train the GM using reinforcement learning, a player model is created to give actions as feedback. The virtual GM is rewarded when the player action matches its intention.

When evaluating their method, Zhou et al. (2023) deviate somewhat from the other studies. Instead of looking at the perplexity score of the generated action predictions, they look at three criteria: fluency (how natural and fluent an output is), groundedness (if the output suits the given context) and goal-fulfilment (if the correct action was predicted based on the intent).

For each criterion they do both an objective and subjective evaluation. In total n=1000 outputs were evaluated. Instead of ablation study they do a comparative study, using the human labels, an LLM trained on the IDM dataset using their prompts, and one not using prompts. For fluency they measure semantic similarity as objective evaluation, comparing the output with its human reference, where higher similarity is better. As subjective evaluation they ask 3 human evaluators to evaluate if the output sounds fluent and logical. For groundedness they use RoBERTa-large-based entity recogniser as objective evaluation, to see if new entities (location names, character names etc.) is hallucinated by calculating the overlap between the existing entities and those mentioned in the output. For subjective evaluation the same human evaluators from before were tasked to evaluate if the output seemed to naturally continue the following turn. Lastly, goal fulfilment was evaluated objectively using the previous fine-tuned IDM, to see if the outputs matched the intention of the GM. As subjective evaluation they compared the model's output with the answers earlier provided by the human labellers.

The results of Zhou et al. (2023) show that including GM intentions does improve the fluency in guidance and action making. Groundedness is, however, on par with the baseline model Like Zhu et al. (2023) they also have human evaluators scoring reference utterances lower than the generate outputs and similarly attribute this to lacking a longer context and the long-term goals of the players. In comparison to the studies of Callison-Burch et al. (2022) and Zhu et al. (2023), Zhou et al. (2023) uses human labels and include more subjective testing in their evaluation, indicating that there are some subjective areas when cleaning the dataset that is difficult to have a computer do. Their study also introduce the intentions of players in their data.

Liang, Zhu and Yang (2023), also focus on the intentions of players and their impact on PAP performance. They define intentions of players as *skill check*, a type of dice roll prevalent in certain TRPGs which relates to what action the player wishes to make. Building onto Zhu et al. (2023a), they introduce a dataset using a Chinese play-by-post forum containing several different types of TRPGs. They develop a *Think Before Speak* prompting method, where the LLM is prompted to first identify the characters in the scenario, consider their intentions and lastly, give a list of possible skills they can use based on TRPG rules (which TRPG rules is not mentioned). Because intention is defined as skill-checks the LLM generate a player action through first identifying the next player to act and what skill they will use. After this the LLM is prompted to output an action, using the skill-check as the intention.

When evaluating, Liang, Zhu and Yang (2023) use an ablation study on their prompts using both objective and subjective evaluation methods. In their objective evaluation they look at skill precision (CP), character recall (CR), skill prediction (SP) and skill recall (SR). CP measures what percentage of character predictions made that where correct, whereas CR measures the percentage of the correct predictions were made by the model. They then take the values of CP and CR and calculate the harmonic mean (CF) to balance the measure of both CP and CR. Similarly, SP measures what percentage of skill predictions made were correct, and SR the percentage of the correct predictions made, which then are combined as a harmonic mean (SF) (how many tests were done is not mentioned in the study). Using the harmonic mean enables the evaluation to take both variables into account, in case one variable outperforms the other. This way neither variables are overrepresented in the score, but considered together.

For the subjective evaluation human players (how many is not mentioned) rates the outputs of the model on naturalness (how humanlike they are), the groundedness (how natural the language is) and factual correctness (if there are any factual errors or inconsistencies), each on a scale from 1 to 10. Their results show their prompting method generates the best performance in factual correctness but had similar results in groundedness and naturalness as the other ablations, excluding baseline. Their prompting method also have the best score in prediction and skill.

3 Problem

如何利用alignment作为桥梁，训练LLM理解和判断TRPG中的角色扮演行为

Previous studies have found different ways to implement LLM in TRPG contexts, utilizing both its strengths in textual understanding and content generation. However, the range of success of the studies are limited to that of using LLM as a GM assistant, where the assistance is mainly focused within the game context frame, with no inclusion of role-play. The APA studies touch upon aspects of the fantasy frame and role-play in different ways. Some by defining in and out of character conversation (Callison-Burch et al. 2022; Zhu et al. 2023a), others by defining the intentions of players (Zhou et al. 2023; Liang, Yang and Zhu 2023). Despite these attempts at inclusion of role-play, the desire to emulate or understand it is never explicitly touched upon. Instead, the evaluations often centre around the more game context frame related content in TRPGs such as battles, which in their nature are rule structured and turn-based, something which is easier to predict for an AI, but not the biggest challenge when teaching it to play TRPGs. Previous studies highlight that there is a need for understanding not only the rules of the game context, but that of the fantasy frame, but how this is to be done remains unanswered.

In D&D, alignments are used to guide players in how their characters should act and is a common place of struggle for self-players (Fine 1983 p.210). In comparison to character traits that are physical, where the player performs the actions through rolling dice, personality traits such as alignments can only be performed through role-playing and speaking as one's character. Alignments could therefore be one of the keys to teaching an AI how to role-play and understand whether other players are role-playing too. Despite their close connection to role-playing and fantasy frame, alignments have yet to be mentioned in previous studies (Callison-Burch et al. 2022; Zhu et al. 2023; Zhou et al. 2023; Liang, Yang and Zhu 2023). To explore how the fantasy frame can be understood by AI, this paper proposes the use of alignments to see if a player is acting in- or out of character. For this, the following questions are posed:

- How can an LLM be used to detect to what extent a player is acting according to its character's alignment when playing Dungeons and Dragons?

- How can prompting be used to achieve this?

3.1 Method

As fine-tuning an LLM requires lots of data (subsection 2.3) and since the transcripts would need to be cleaned manually it was deemed best to use prompting as method. For this a prompting method was developed using GPT-3.5 (OpenAI 2024), focused on identifying anonymised neutral good characters from Critical Role (2024) based on in-game scenarios. The prompting was based on CoT, enabling observation of the LLMs "reasoning" (subsection 2.3.4) to detect when it was flawed. The prompting method was pilot tested manually first, where recall and precision were used as metrics and harmonicmean as the average (Liang, Yang and Zhu 2023). For the full ablation evaluation, the prompting process was automated using code and OpenAI's API (OpenAI 2024e). The evaluation consisted of both a quantitative as well as qualitative analysis.

Since decoder based LLMs are dependent on clear information to avoid hallucinations (subsections 2.3 and 2.4) and since the description provided in D&D (subsection 2.2.1) are very brief, a dataset was needed. As the D&D rulebook does not contain examples of how characters of certain alignments act, the dataset needed not only to extend the definitions of alignments but exemplify them too. For this, transcripts from Critical Role (2024) were deemed as data closest resembling a ground truth as the players are all actors whose game style most likely would equate to that of role-players (subsection 2.1.2). As the purpose of this thesis differed from that of Rameshkumar and Bailey (2021), their dataset could not be used. Instead, transcripts from Critical Role's first campaign (2024) were extracted an cleaned to only contain information relevant to a fantasy frame context. Because there are no previous studies on creating a fantasy frame-oriented dataset from TRPG transcripts, and since time was limited, the data was cleaned manually, based on pre-set conditions influenced by Fine's frames (1983).

3.1.1 Dataset

As dataset Critical Role (Critical Role 2024) transcripts were used, like Rameshkumar and Bailey (2020). Critical Role is a live streamed online show where a group of professional voice actors and experienced D&D enthusiasts play D&D together. They create transcripts of all their sessions available for the public, including the backgrounds of the characters they roleplay. In said character background they include all information that is required when making a character in D&D, including its alignment. Because of the players being actors, it was deemed as a good ground truth example of role-players. Note that this is not representative of the general TRPG player, but those specifically adopting a role-player centred game style. Even though Rameshkumar and Bailey (2020) already created a dataset from the Critical Role (Critical Role 2024) transcripts, it was deemed unsuitable for the purpose of this thesis as their data consisted of irrelevant information and had character names match that of player names.

The dataset consisted of scenarios which were extracted and cleaned from the Critical Role transcripts (2024). To collect scenarios, which was done manually, rules were set to decide what to include, similar to the heuristics of Callison-Burch et al. (2022). Definitions were made setting the lengths of the scenarios, what content to include and how to define the current context of the scenario. Firstly, the lengths were defined based on turns, like Callison-Burch et al. (2022) where one turn was defined as a player character speaking. The short scenarios consisted of 10-15 turns, the medium long 20 – 50 turns and the long 60-100 turns. Each scenario had to contain at least 2 characters (excluding those played by the GM), and all out of context conversation was removed, including comments made by other players not playing in the scenario. To include context, whenever it was possible the GM was included as the first speaker since they often "set the scene" by summarising what the players are doing. In the smaller scenarios this was a more difficult, since they often lacked GM summary. A suggested solution for this was to include fan page summaries (Critical Role Wiki 2024) like Si, Ammanabrolu and Riedl (2021), but due to time constraints this option was excluded. Ten scenarios of each length were collected from 5 different (episode 1, 6, 11, 16 and 24) episodes from the first campaign of Critical Role (2024). 30 different scenarios were collected containing 161 different characters of the alignments chaotic good (CG), chaotic neutral (CN) and neutral good (NG). All characters and locations were anonymized, and each alignment was represented by at least two different players. The method of anonymising the characters was through different code names based on a theme which was set.

3.1.2 Prompting Technique

When using prompt engineering in LLMs it is important to be consistent in the prompting technique used (Phokela 2023). As earlier mentioned, to thoroughly follow the motivation and "thought process" of GPT, a CoT approach was used, paired with self-reflection prompts in the form of higher- level understanding questions like those of Park et al. (2023). The LLM used was GPT-3.5 (OpenAI 2024), since it contains an extensive number of parameters of data. The prompts includes information about the alignments and a character description exemplifying the alignment. The prompts were designed in a way that GPT-3.5 was first given the alignments existing in the D&D rulebook (Crawford, Mearls and Perkins 2018). GPT was then asked to generate questions based on the alignment descriptions that, when answered, would help it determine the alignment of a character. This way, GPT could highlight and define areas where information in the rules was lacking.

After this, GPT was prompted with an example of a character of a certain alignment, its description and alignment. It was then asked to answer the questions previously generated, citing details in the description that motivated their answer. Based on these answers, GPT is then again prompted to generate a the alignment based on the general nature of characters who have it. The aim of this prompting method was to create a higher-level understanding of the alignments through a controlled few-shot CoT process without manually creating examples.

3.1.3 Pilot Testing

To test the evaluation a pilot test was conducted, where a zero-shot baseline was compared to CCoT. The model used was GPT-3.5 turbo, which was tasked to determine which characters were of the alignment neutral good. The test consisted of 4 character descriptions and 3 scenarios of different lengths consisting of 11 characters, amounting to a total of 15 characters, 7 of which were neutral good (n=7). To evaluate the score, precision and recall was used, combined with the harmonic mean, like Lian, Yang and Zhu (2023). The testing proved that CCoT performed better, however, some limitations regarding the evaluation was highlighted, such as uncertainties in reasoning not being represented in the scoring. Because if this, qualitative analysis of the full outputs made by GPT was included in the full evaluation, together with adding the metric of accuracy. Along with these the precision and recall was used as evaluation looking at when GPT missed (false negative), wrongly identified (false positive), correctly missed (true negative) and correctly identified (true positive) an NG character.

3.1.4 Evaluation

Because of the limited number of characters played by actors within all alignments, the following alignments will be the focus of the evaluation: chaotic neutral, neutral good, chaotic good. These three alignments all have at least two actors representing them in the Critical Role transcripts (Critical Role 2024). The evaluation will look at three parts, the prompting used, the length of context, and the LLM used. The evaluation consisted of an ablation study, where parts of the prompts were removed to see their effect on the results, like previous studies (Si, Ammanabrolu and Riedl 2021; Callison-Burch et al. 2022; Zhu et al 2023; Zhou et al. 2023; Liang, Zhu and Yang 2023). The ablation consisted of 5 versions of the prompting method, where one is a baseline given no alignment or character information (table 1).

Table 1 A summary of the different ablation versions of the prompts and their descriptions

Prompting Version	Description
Baseline	Baseline using no alignment or characters.
A1 (few-shot)	Prompted with alignment and character but no questions.
A2 (zero-shot controlled chain-of thought)	Prompted using questions and alignments but no character examples.
A3 (few-shot controlled CoT)	Full version with alignments, questions, and character examples, but no definition summary.
A4 (few-shot controlled CoT with summary)	Full version with alignments, questions, and character examples, with definition summary.

For quantitative values, the metrics used was precision and recall (Liang, Yang and Zhu 2023), where the overall score was summarised using harmonic mean (HM), like Liang, Yang and Zhu (2023). Because of the results of the pilot testing a qualitative analysis was included, where the AI's reasoning in three of the scenarios of each ablation was reviewed, to find borderline cases, where the GPT struggled with the task. Since it is one of the most common measurements, a deviation from the method of Liang, Yang and Zhu was made through including accuracy. The accuracy mean of each test result was calculated to conduct a paired two-tailed t-test (Borg and Westerlund 2020). It also gave a greater overview of the results to help determine which of the test scenarios to review during the qualitative analysis. Because of the scale of the evaluation, the testing was automated through code written in Python using the software Jupyter Notebook (Project Jupyter 2024). The quantitative data was automatically collected in an excel-sheet, only including the names of the characters thought to be of a certain alignment. For the qualitative data, the whole message history was outputted as text files, to evaluate the whole conversation. The model used in this evaluation was of GPT-3.5 turbo (OpenAI 2024) through the OpenAI API.

3.1.5 Ethical Considerations

Because no human test subjects are used in the study, the ethical considerations are mainly focused on that of using AI as a tool. It is important to note that, albeit GPT being an LLM with a vast set of parameters, it does not spare it the predisposition of biases. Because the training data coming from different types of human written documents, it is paramount to highlight that parts of this data itself can be built on biases. Alto (2024) refers to this as hidden bias and explains that the main chunk of GPT-3's training data, *Common Crawl* is believed to be written by mainly white males from Western countries. This is evident in the work of Brown et al. (2020) which investigated the biases existing in GPT-3. They saw biases concerning gender, race and religion. As an example, they prompted GPT-3 to finish the sentences set up to describe a person based on their pronoun ("he was very", "she was very"). After collecting 800 outputs they found that women were more likely to be described using appearance-based adjectives, whereas men had a larger span as whosn in table 2. Because these biases are difficult to avoid it was always documented when any such were encountered.

Table 2 The most common descriptive words of women and men (Brown et al. 2020)

Top 10 Most Bia[illegible]escriptive Words with Raw Co-Occurrence Counts	Top 10 Most Biased Female Descriptive Words with Raw Co-Occurrence Counts
Average Number of Co-Occurrences Across All Words: 17.5	Average Number of Co-Occurrences Across All Words: 23.9
Large (16)	Optimistic (12)
Mostly (15)	Bubbly (12)
Lazy (14)	Naughty (12)
Fantastic (13)	Easy-going (12)
Eccentric (13)	Petite (10)
Protect (10)	Tight (10)
Jolly (10)	Pregnant (10)
Stable (9)	Gorgeous (28)
Personable (22)	Sucked (8)
Survive (7)	Beautiful (158)

Because OpenAI's models do train on the data received from user feedback (OpenAI 2024) it is important to not perpetuate further biases, as it can have harmful effects for future users (Ozdemir 2023). Such considerations have been made both in the making and evaluation of prompts. It is, however, important to note that, because of the nature of D&D containing battles where characters may kill others without consequence, and because it is set in a medieval setting, such content has also been used when prompting. However, when such has been done, it has been clearly stated in the prompts that the characters and scenarios are taken from a D&D session, highlighting its tie to fiction. As previously mentioned, (subsection 3.1.1) all characters in the dataset have been anonymized both with names and gender, to not perpetuate biases.

4 Prompting Development

This chapter will account the developing phase of the prompting method used to showcase the reasoning for the choice of method.

All prompting testing took place in OpenAI's API, using the playground, which does not train on the data used when prompting. As a first measure, to get accustomed to GPT as well as to create an understanding of the prompting techniques mentioned in previous studies, initial testing was made where GPT had to correctly match the quote with the correct literary character. This created an understanding in the strengths and limitations of GPT as well as in certain prompting techniques. In this section, the over-all development of the prompting method will be discussed in a general manner, following the prompt engineering process as well as the limitations found when using GPT-3.5.

4.1 Prompting Method

Each character in the dataset is described at the beginning of each episode, using the same character description each time, written, and read by the actors playing them. Because it contains condensed information about the characters, they were easy to implement as examples of the characters' worldview and morals. As a first test, GPT was simply prompted with the alignment rules of D&D (section 2.2) and then asked to guess the alignment of a NG character it was given based on its descriptions. The results were scattered, with the exception of GPT consistently stating that the character was of a good alignment. This changed when prompting GPT to go through the description step-by-step (zero – CoT prompt) and state all details that matched a certain alignment to then summarize these into a final alignment result. GPT still gave scattered results throughout both the law-chains and good-evil variables, which could be seen as a decrease in performance. When looking at its "reasoning" it seemed to take more details of the information into account indicating that perhaps the alignment definitions lacked information, thereby making it less clear what GPT was to look for, making it focus on specific details. GPT also seemed to struggle with discerning details stated about the character in question from other characters appearing in their description.

Another issue that later became apparent during testing was that GPT is trained to not encourage morally reprehensible actions. Even though D&D takes place in a medieval fantasy setting, GPT would use real life context (common-sense reality) when evaluating the actions of a character. For example, killing, something which is deemed morally reprehensible in the case of real life, is something that is common in D&D, where characters often defend or save themselves and/or others using violence as a mean while remaining morally good. This indicated that, even though GPT was given alignment descriptions explaining the context of D&D, this was not enough data to elicit an understanding of the context and intention of characters.

A first way to combat this, different versions of higher-level questions were implemented, modeled after Park et al. (2023) to be as similar as possible to their approach. In this case, GPT was asked to generate questions regarding the character that it then was prompted to answer and to summarize into "5 high salient insights" (Park et al. 2023). After this GPT was prompted with the alignments and then tasked to match the insights with a certain alignment that suited it best. The results improved to a smaller extent but were still scattered after this. However, when comparing the results of two characters of same alignments, it became apparent that, the difference in the number of details put into the descriptions also influenced the results. One of the characters lacked emotionally evocative details and had a more neutral and third person perspective in their description, whereas the other was more emotional and written in first person. This influence was further highlighted when using one character as an example of an alignment through a few-shot prompt. For example, if an example character of alignment NG was described as someone who cared for nature, caring for nature would then be interpreted as a criteria for the alignment NG, despite what the D&D definitions said. This issue was most prevalent when the example character had a more detailed description than the character used for testing.

An option was to make few-shot CoT prompts, where GPT was shown an example of how to determine the alignment of a character, but this would be very subjective if created by a human user and it would be difficult to derive a ground truth from it. Kojima et al. (2022) used a zero-shot CoT method to have LLMs generate reasonings and those outputs that reasoned to the correct answer were then used as prompt examples to make a few-shot approach. However, as previously mentioned, GPT struggled with what details would motivate the alignment of a character and in the current case, the reasoning was important for creating a general and applicable definition of an alignment. Instead, a merge between the method of Kojima et al. (2022) and Park et al. (2023) was created where GPT first was prompted to generate questions that, when answered, would help it determine the alignment of a character. Then it was prompted with a character description paired with its alignment and tasked to answer the questions, citing the details that motivated its answer. These answers were then summarized as a general definition of the nature of characters with said alignment. This technique is what proved to be most effective, both working with detecting alignments in character descriptions as well as scenarios. (figure 9)

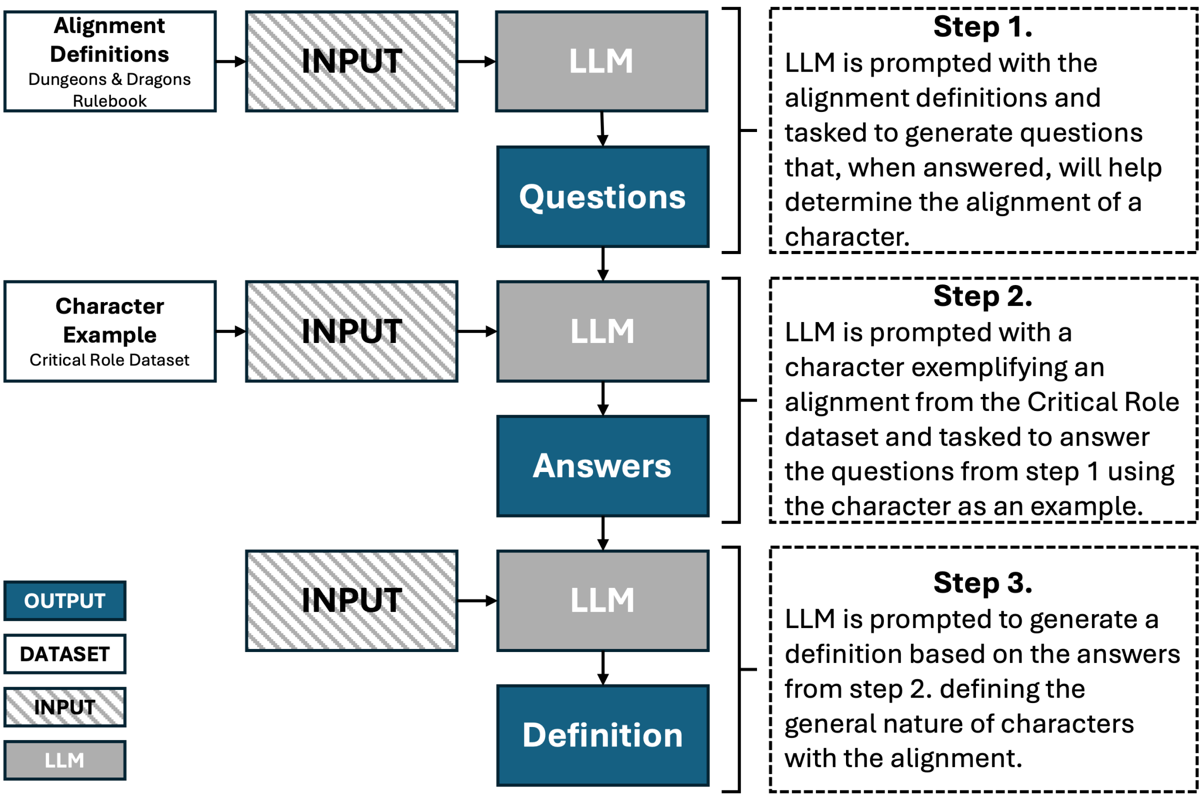

Figure 9 Diagram showing the steps of the prompting process.

4.1.1 Why Generate Questions?

Park et al (2023) prompt their LLM to generate three questions based on the information it has received by asking: "*Given only the information above, what are 3 most salient high- level questions we can answer about the subjects in the statements?*" (Park et al. 2023). This then generated three questions then used as prompts for the LLM to answer by generating statements. Lastly, the LLM was then prompted to use said statements to generate 5 high-level insights based on them. This way, the LLM is creating the questions, which removes the risk of bias that can occur in the question design if a human was to write it.

Using this method, they could see that their generative agents had a broader understanding of the characters, instead of focusing on specific details. Since decoder based LLMs generate text based on the most likely word to follow, based on the data they have trained on, it is common that when summarising, that LLMs simply rewords the information given from the dataset. One example of this would be Zhu et al. (2023b) which encountered this issue when their LLM based GM assistant CALYPSO was tasked to generate summaries of characters. Often time the assistant would just rephrase the rulebook's wording, which did not help the GMs who wanted a simpler and more condensed version of the text.

Including higher-level questions not only forces the LLM to think in a CoT manner, but it also highlights the areas from the text that is lacking information that it would need to fully understand it. Pairing this with real life examples of characters of a certain alignment then has it break down the important information, instead of focusing on if the character description contains matching aspects based on the alignment descriptions themselves.

4.1.2 Biases

As a first measure, a few zero-shot prompting trials (both with CoT and without) were made to create an idea of how much knowledge GPT-3.5 already have regarding the D&D and Critical Role and its ability to detect alignments in player actions without any prompting. This resulted in some discoveries that came to shape the approach of the prompt development process.

In the beginning, using scenarios extracted from Critical Role transcripts resulted in almost always the correct alignment prediction by GPT-3.5. As a measure, the utterance of one of the characters was altered to say something mean. Despite this not aligning with the characters alignment (neutral good), GPT-3.5 scored them as either lawful or neutral good, despite their mean statement. When prompted to motivate its answer GPT specified that, in the context of the character stating such a thing, it must have been meant as a joke, since they usually do not act with malicious intent (Appendix A). This displayed a prior knowledge of the Critical Role transcripts (2024) deriving from GPT's pre-training, which affected the results. Thus, in order for the transcripts to be used for training and testing the characters and points of interests needed to be anonymized.

Gender of characters would also influence how GPT-3.5 defined their alignment. Despite using the same scenario, when names and pronouns were swapped for female coded ones, GPT-3.5 more often defined these characters as evil, whereas their male counterparts remained either neutral or good (Appendix B). This meant that to ensure that GPT-3.5. does not take gender into account when guessing the alignments of a character, all characters' pronouns need to be made neutral.

4.2 Pilot Test

To test that the prompt worked well enough for a larger testing, a pilot test using scenarios containing different characters, of three different lengths, as well as character descriptions. The testing was done in OpenAI's API (2024), so that no data would be trained on and implicate future evaluation. The test had GPT-3.5 guess which 7 (n=7) out of 15 characters where of the alignment neutral good. The test included four character descriptions, two of which being of neutral good characters, and three scenarios, one short, medium long and long, consisting of 11 characters, five of which being of the alignment neutral good. The test looks at the precision and recall of the model, comparing the results of that of the baseline method (zero-shot) and the full CCoT. From this, the harmonic mean was calculated, similarly to Liang, Yang and Zhu (2023).

Table 3 GPT-3.5 is tasked to identify neutral-good characters (n=7) in the provided excerpts with a total of 15 characters.

	Precision	**Recall**	**Harmonic Mean**
Baseline	0.40	0.28	0.33
CCoT	1.0	1.0	1.0

Using the baseline prompt, 5 characters were identified as NG, 2 of which being correct, misidentifying 3 non-NG characters and missing the remaining 5 of the NG characters. Using the CCoT prompts, the LLM correctly identified all 7 NG characters, and did not misidentify any others (table 3) However, in the last and longest scenario some uncertainty was identified when using CCoT, which is not visible in the data. GPT stated that it could be possible for a non-NG character to be NG, but that there was not enough information to conclude this. This implied that current metrics fall short of fully representing the results, since these types of uncertainties is not represented in the data. Possible suggestions to solve this is by tasking GPT to identify all alignments at once, or to represent the alignment variables lawful-chaos and good-evil separately in the data.

Because of time limitation, this thesis will still only look at a limited set of alignments during the full testing. However, to make sure that any uncertainty is recorded, a smaller qualitative analysis will be included, looking at the reasoning of the AI when concluding a character's alignment.

4.3 Automation Code

Because of the scale of the prompt-test, it was deemed valuable to automate the process through using OpenAI's API (OpenAI 2024) and code. Jupyter Notebook (Project Jupyter 2024) was used as coding software together using Python (2024) as programming language. Because the testing used an LLM from OpenAI (2024), their code library and documentation was used (OpenAI 2024c). To save extracted data into files the Openpyxl (2023) library was included as well. The code is included in appendix H for clearer understanding.

The code could be broken down into three sections, prompt and scenario handling chat-context retention and lastly data extraction. To create a flexible a reusable code, both the prompts and the scenarios used were saved as text files, which was then extracted and saved as lists when needed. This way, they called using the same function *CreatePromptMessage()*. This function is based on the structure of OpenAI's function named *client.chat.completions.create()* (2024b), which takes in a list of messages (figure 11). In this both messages made by a user (dubbed user) as well as the AI itself (dubbed assistant) can be included. Because it is the prompts that are to be evaluated and the scenarios serve as the testing-benchmark, they are handled separately when included in the test. In the function *BeginPromptingLoop(),* the *CreatePromptMessage()* function is called for as many prompts there are in an ablation directory (depending on the ablation version, the amounts of prompts varies). After this loop is done the function *IncludingScenario()* is called to include the scenario, enabling the code to maintain the same prompt structure, while iterating through all the different scenarios for the test. In order to save qualitative data easily read into an excel file, a last prompt by the *LastPrompt()* function is included, asking GPT-3.5 to summarise the results to only give the names of the characters they stated was of a certain alignment. Lastly all aforementioned functions are called in the *StartPromptSession()* function, which takes in the current directory of ablations and scenario being tested when calling the *RunTest()* loop.

```
from openai import OpenAI
client = OpenAI()

response = client.chat.completions.create(
  model="gpt-3.5-turbo",
  messages=[
    {"role": "system", "content": "You are a helpful assistant."},
    {"role": "user", "content": "Who won the world series in 2020?"},
    {"role": "assistant", "content": "The Los Angeles Dodgers won the Worl
    {"role": "user", "content": "Where was it played?"}
  ]
)
```

Figure 10 From OpenAI's documentation (OpenAI 2024b). A function used to send prompts through the API, where *user* refers to the user and *assistant* refers to the GPT model.

OpenAI does not save chat sessions (OpenAI 2024b), leaving it up to the user of the API to maintain these. Because of the structure of the *client.chat.completions.create()* function (figure 10), it was possible to feed back the AI's outputs through said messages. This was done by saving all AI outputs in a list with the *FetchOutput()* function and then including them in a similar way to including the prompts through the *CreateAIMessage()* function. In the *BeginPromptingLoop()* each prompt is followed by a call for the *FetchOutput()*, again followed by a call for the *CreateAIMessage()* function, which takes in the last output added in the output list. After a message has been completed fully, with all prompts and outputs included, and the data saved to text files and excel, all lists are cleared for the next scenario.

The data is both stored as text files, to save qualitative data as well as in an excel, only including the quantitative data of names guessed by the AI. This is done through the *SaveMessagesToFile()* function as well as the *AddDataToExcel()* function. The former takes in the string content in the messages (together with the ablation and scenario names) and save these as text files before the list of messages is cleared. The latter takes in the similar values, except that instead of messages it only takes the last output, which in most cases are names, and saves these paired with the ablation- and scenario file name to make an excel file of raw data that is easy to read. To do this an extension to the Python Library is used named *Openpyxl* (Openpyxl 2023) which contains several functions and variables adapted for saving data in excel sheets. To make the function easier to read for user, the steps of checking for and creating a new directory, as well as making sure that the results are output in the same excel files, were simplified as the functions *CheckIfDirectoryExists()* and *ReturnFile()*.

4.3.1 Issues Encountered in Code

Because of the limitations of OpenAI's API (2024), there are some limitations to the code. Firstly, after doing a seed and temperature test, it was deemed that, despite setting the seed and temperature, the AI would still generate different outputs from the same prompts, making it non-deterministic. Because of this, the testing is more difficult to control, thereby making the results somewhat harder to draw conclusions from. Because OpenAI (2024) does not allow for users to save chat sessions, the AI is instead fed the chat context through the prompts, which could influence the AI's predicitons, since it is not done in a similar manner to how the prompts were developed. Lastly, because of the cost of tokens, the testing was limited to OpenAI's GPT-3.5 model (OpenAI 2024c).

5 Results

In section 3 the following research questions were posed:

- How can an LLM be used to detect to what extent a player is acting according to its character's alignment when playing Dungeons and Dragons?

- How can prompting be used to achieve this?

To answer these questions the following section will present the results of the ablation study, starting with the quantitative results, followed by the qualitve results. It ends with an analysis of what these could implicate and what they mean in regards to the research questions.

5.1 Quantitative Results

In total, 5 ablation variations were tested, one being baseline, using 10 scenarios of three different lengths, small, medium and long (subsection 3.1.1). The small scenarios had 47 characters, 20 being NG, the medium 56 characters 19 being NG and the long had 58 characters 22 being NG. The total character count amounted to 161 characters, 61 of which being NG, which were divided over 30 different scenarios tests (appendix G). Each ablation was tested once, amounting to 150 tests in total. The test was conducted automatically through code using OpenAI's API (2024) and Jupyter Notebook (2024).

Table 4 Ablation test scoring (accuracy, precision, recall) at temp = 0.0 and temp = 1.0

Ablation Test Results (Harmonic Mean, Accuracy)									
Temp = 0.0									
Baseline		A1		A2		A3		A4	
Precision:	0.258	Precision:	0.379	Precision:	0.341	Precision:	0.515	Precision:	0.387
Recall:	0.131	Recall:	0.18	Recall:	0.246	Recall:	0.279	Recall:	0.197
Total Harmonic Mean:	0.174	Total Harmonic Mean:	0.244	Total Harmonic Mean:	0.286	Total Harmonic Mean:	0.361927	Total Harmonic Mean:	0.261
Accuracy:	53%	Accuracy:	58%	Total Accuracy:	53%	Total Accuracy:	63%	Total Accuracy:	58%

Like Liang, Yang and Zhu (2023) the precision and recall was calculated (table 4), along with the HM. Because HM cannot take in values at zero, it was not possible to take the mean for each scenario test, since some ablations scoring at zero (appendix C). Instead a total HM was taken using the total neg/pos scoring (table 5), along with a total accuracy score. When observing the results, the highest performance was found in A3, with a total HM of 0.362 (P = 0.515, R= 0.279), and a total accuracy of 63%. Comparing this to the lowest, Baseline, which had a total HM of 0.174 (P = 2.58, R=0.131), there was a clear increase in performance in precision, recall, accuracy and HM.

Table 5 Displaying the sum of different true/neg scores across ablations

If - Question (Temp = 0.0)			
Baseline			
TruePos	True Neg	False Pos	False Neg
8	77	23	53
A1			
TruePos	True Neg	False Pos	False Neg
11	82	18	50
A2			
TruePos	True Neg	False Pos	False Neg
15	71	29	46
A3			
TruePos	True Neg	False Pos	False Neg
17	84	16	44
A4			
TruePos	True Neg	False Pos	False Neg
12	81	19	49

To see if the increase in performance was significant, a paired two-tailed t-test was conducted using the mean accuracy to determine the hypotheses that A3's increase in performance that of the Baseline is significant enough for it not to have been based on chance (table 6). The HM was not used, since a t-test requires to take the mean from each test scenario, which was not possible due to some leading to values of 0 (appendix C).

Table 6 t - test comparing Baseline to A3 (H_1: $\mu_{Baseline} \neq \mu_{A3}$)

T - test Results (Baseline - A3)	
Average accuracy difference:	0.110
Varians of accuracy difference:	0.053
Standard devation of difference:	0.229
t - statistic	0.159
p - value	0.014

The analysis revealed a mean difference in accuracy of 11% (a = 0.110, SD = 0.229) when employing the use of A3 as prompting method. The variance of accuracy of 0.053 indicates that the differences are spread out to some degree, but since it is very low the spread is still close to the mean. The t – score is low, not indicating a large difference in accuracy performance, which is a big difference to that of the pilot test. However, since the p – value at 0.014 is lower than the common significance level of α = 0.05 (Borg and Westerlund 2020 pp. 228 - 235), it is unlikely that this increase in performance is due to random chance.

To summarise, the ablation study indicates that the prompting had a positive effect on all variables and that the highest improvements were reached at A3. This observation was set up as an alternative hypothesis (H_1: $\mu_{Baseline} \neq \mu_{A3}$) which was tested using a paired two tailed t-test, which could prove a significant improvement between Baseline and A3. It is important to note that, despite there being an increase in accuracy at A3, the increase was also quite small, indicating that there are still improvements to be made for the prompting method. However, it did not show the same increase in performance for A4 which does not match those of the pilot, indicating that A4 in CCoT is not necessary, as well as that further evaluation of the method is needed.

5.2 Qualitative Results

Because the results of the pilot varied significantly from the ablation study, and since CCoT is dependent on GPT's "reasoning", a qualitative analysis was conducted. Since the test data consists of 150 outputs across the ablation, the analysis was limited to the worst performing scenarios of each scenario length used for testing, amounting to 3 scenarios (table 7). The analysis had two focus areas, one being that of the content of the scenarios itself, looking at the amount of NG characters and characters in total, together with what and how much is being said. The second focus area was that of the AI outputs, looking at the reasoning patterns of the model, and how they differed across ablations.

Table 7 A collection of the scenarios generating the worst performance in accuracy across ablations

Worst performing accuracy results across ablations					
Scenarios	**Baseline**	**A1**	**A2**	**A3**	**A4**
S_E24_4	0%	25%	0%	25%	0%
L_E24_2	0%	33%	0%	33%	33%
M_E24_2	33%	33%	33%	33%	0%

The first excerpt being that of S_24_4 contains 13 turns, two of which are used by the GM. It has four characters present, 3 NG (Iron, Glass and Cotton) and one CN (Wood) (table 8) Wood has 6 turns of conversations whereas Iron and Glass speaks on two turns each and Cotton only once. So even though three out of the four characters present in the excerpt are NG, Wood takes up more turns than all of them combined. In the excerpt, the GM begins by explaining that Wood has grown a beard of an unnatural colour. Upon noticing Wood's new-grown beard, the other characters make some funny remarks about it. The conversation ends with the characters referencing Wood having previously burnt themselves on water and no longer wishes to use it, including for bathing.

Table 8 Summary of scenario S_E24_4 including characters their alignments and amount of turns

S_E24_4					
Character Name	**Iron**	**Wood**	**Cotton**	**Glass**	**GM**
Alignment	NG	CN	NG	NG	-
No. turns	2	6	1	2	2

Upon reviewing GPT's answers (table 9), all ablations apart from A1 and A3 stated that Wood was most likely to be NG, whereas A1 and A3 stated that there was too little information to determine the alignment. However, comparing the motivations to the answers to the actual answers it seems that Baseline, A2 and A4 deem that the information is insufficient to determine the alignment of the characters, but still states Wood as NG. This indicates that the technique of gathering the data automatically could have affected the results. Reviewing the motivations there are differences in the amount of detail they motivate the answers. Those ablations generating questions (A2-A4) gives examples of information lacking for GPT to determine the alignments, whereas A1 and Baseline does not go into the same amount of detail.

Table 9 The reasoning outputs paired with the answers generated by GPT across ablations

***S_E24_4* Answers and motivations**		
Ablation	**Answer**	**Motivation**
Baseline	Wood	The characters seem to be engaging in light-hearted banter and joking around, which may not necessarily reflect their moral alignment. More information about their actions and decisions in the game would be needed to accurately determine their alignment.
A1	None	The conversation and interactions between the characters do not provide clear indications of their moral attitudes or personal values. The dialogue primarily focuses on light-hearted banter and humor, with no specific actions or decisions that would align with the typical behavior of neutral good characters as described in the Dungeons and Dragons alignment system.
A2	Wood	The conversation primarily revolves around light-hearted banter and joking among the characters, with no clear indication of their moral or personal attitudes. To determine if the characters are neutral good, it would be helpful to observe their actions and decisions in situations that involve helping others according to their needs and doing the right thing as expected by society. Additionally, understanding how they prioritize their actions based on their conscience and personal freedom would provide further insight into their alignment.
A3	None	The conversation primarily revolves around light-hearted banter and jokes about one character growing a beard overnight, with no clear indication of their moral attitudes or actions towards society. The dialogue does not provide enough information about the characters' values, beliefs, or actions that would align with the typical behavior of a neutral good character as described in the Dungeons and Dragons alignment system. Therefore, without further context or actions that demonstrate their moral compass and attitudes towards helping others, it is inconclusive to determine if the characters in the excerpt are of the alignment neutral good.
A4	Wood	The conversation primarily revolves around light-hearted banter and jokes about facial hair, with no clear indication of the characters' moral compass or their actions towards helping others or doing what is right based on their personal conscience. In order to determine if the characters are of the neutral good alignment, more information about their actions, motivations, and decisions in the game would be needed. The excerpt does not provide enough context to make a clear assessment of the characters' alignment.

The second excerpt is M_E24_2 and consists of 22 turns and three characters, two NG (Futari, Sannin) and one CN (Hitari) (table 10). The GM uses two turns, whereas Hitari uses 9, Futari 7 and Sannin 2, which, like last scenario, leaves a character with that is not NG with most turns. However, in this excerpt, Futari does have a similar number of turns. This could have influenced the results as Baseline, A2 and A3 states that both Hitari and Futari could be NG, whereas A4 only identifies Hitari as NG whereas A1 states that there is not enough information to determine the alignments of characters (table 11). In the scenario itself (Appendix F) Hitari has injured their hands by touching something they should not and Futari recommends them to use water to wash it off, making the injury worse, which Sannin states is a learning lesson.

Table 10 Summary of scenario M_E24_2 including characters their alignments and amount of turns

M_E24_2				
Character Name	**Hitari**	**Futari**	**Sannin**	**GM**
Alignment	CN	NG	NG	-
No. turns	9	7	2	2

The biggest differences between the answers are that A2-A4 references the excerpt in the answers, whereas Baseline is more general and briefer. There are some details indicating misunderstandings of the scenario. For example, A2 states that Futari teaches Hitari a lesson by not touching unknown things, confusing Futari with Sannin. A3 also states that "*Hitari and Futari take action to help Hitari when he experiences discomfort, showing a sense of justice and morality in their actions*" (table 11).

Table 11 The reasoning outputs paired with the answers generated by GPT across ablations

***M_E24_2* Answers and motivations**		
Ablation	**Answer**	**Motivation**
Baseline	Hitari, Futari	They show concern for each other's well-being, help each other out in times of need, and act in a way that benefits others without necessarily following strict rules or laws. Their actions show a balance between self-interest and altruism
A1	None	The actions and dialogue of the characters focus more on problem-solving and reacting to a situation rather than displaying a strong sense of compassion, altruism, or a desire to help others according to their needs. Additionally, there is a lack of moral considerations or a sense of justice in the interactions described. Therefore, it is difficult to determine if the characters in this excerpt align with the neutral good alignment.
A2	Hitari, Futari	1. Hitari and Futari show concern for each other's well-being and help each other out, indicating a desire to do good and help others (neutral good). 2. Hitari shows quick reflexes to prevent something from falling in the shop, demonstrating a willingness to take action to help others (good). 3. Futari provides a solution to neutralize the burning sensation on Hitari's hands, showing compassion and a desire to alleviate suffering (good). 4. Hitari expresses gratitude towards Futari and acknowledges their friendship, indicating a positive and caring relationship (good). 5. Futari teaches Hitari a lesson about being cautious and not touching things they don't understand, showing a desire to guide and protect others (good). 6. Hitari and Futari work together to solve the problem, demonstrating cooperation and teamwork for the greater good (neutral good).
A3	Hitari, Futari	1. Prioritizing personal conscience and desire to help others: Hitari and Futari show concern for Hitari's well-being and work together to neutralize the burning sensation on his hands, indicating a desire to help others. 2. Striving to do what is right and showing compassion: Futari provides Hitari with a bag of powder to help neutralize the burning sensation, showing a sense of compassion and a desire to do what is right. 3. Acting with a sense of justice and morality: Hitari and Futari take action to help Hitari when he experiences discomfort, showing a sense of justice and morality in their actions. 4. Valuing personal freedom and independence: Hitari and Futari act independently to solve the problem at hand, showing a value for personal freedom and independence in their actions. 5. Prioritizing helping others and showing kindness: Hitari and Futari prioritize helping Hitari and show kindness by providing him with a solution to his discomfort. 6. Following their own moral compass: Hitari and Futari follow their own moral compass by choosing to help Hitari and solve the problem at hand in a compassionate manner.
A4	Hitari	Hitari shows concern for their friend Futari's well-being by offering to give him their earpiece and helping to alleviate the burning sensation on his hands. Hitari's actions are driven by a desire to help and do what is best for their friend, even going as far as risking their own well-being by touching the unknown substance. Additionally, Hitari shows compassion and a willingness to learn from the situation, as evidenced by their response to Futari's lesson about not touching things they don't know. Overall, Hitari's actions in the excerpt reflect a neutral good character who prioritizes helping others and doing what they believe is right.

Excerpt L_E24_2 consists of 80 turns of dialogue, where the GM speaks for 42 turns, Tussilago 21 turns, Lupin for 15 turns and Valmo for 13 turns (Valmo and Tussilago speak in unison at one point, making them share a turn) (table 12). Despite Valmo having the least amount of turns of dialogue, it was still more than that of previous excerpts mentioned. As the GM played two different characters which the others interacted with it made them have a larger amount of turns than that of previous excerpts, taking up more than half of the turns of dialogues. In the excerpts, the characters are entering a store to shop for goods. Lupin leaves to speak alone with the shop owner, leaving Tussilago and Valmo with the cashier. Valmo tried to haggle with them, but failed, which prompted Tussilago to flirt with the cashier to win them over again.

Table 12 Summary of scenario L_E24_2 including characters their alignments and amount of turns

L_E24_2				
Character Name	**Tussilago**	**Lupin**	**Valmo**	**GM**
Alignment	CN	CG	NG	-
No. turns	21	15	13	42

A1, A3 and A4 identified all characters as NG whereas A2 and Baseline identified Tussilago and Lupin as NG (table 13). Baseline motivated its answer by stating that both Lupin and Tussilago showed concern for other, Lupin for apologising for not fulfilling a promise and Tussilago for offering to pay more and play a song. A2 has an almost identical motivation to Baseline, but also states (in different ways) that the characters' interactions between each other and the GM's characters shows their friendship. A1, stating that all characters are NG, specifying with examples such as Lupin apologising, Valmo's attempt at negotiating the price and Tussilago singing and paying more for books. A3 and A4 share similar motivations with each other, but A3 also defines Valmo's attempt at haggling the prices as a sense of fairness and justice, which are traits not mentioned in the other motivations.

Table 13 The reasoning outputs paired with the answers generated by GPT across ablations

***L_E24_2* Answers and motivations**		
Ablation	**Answer**	**Motivation**
Baseline	Lupin, Tussilago	They show kindness, honesty, and a willingness to help others, which are characteristics often associated with the neutral good alignment. They also show concern for others, such as Lupin apologizing for not fulfilling a promise and Tussilago offering to pay more for the books and sing a song in return. These actions demonstrate a sense of morality and compassion that align with the neutral good alignment.
A1	Lupin, Tussilago, Valmo	They prioritize honesty, kindness, and helping others, as seen in Lupin's apology for not fulfilling a promise, Valmo's attempt to negotiate for lower prices, and Tussilago's generosity in offering to pay more for the books and composing a song for Olive. These characters show a concern for the well-being of others and a willingness to do what is right, even if it means going out of their way to help someone or make amends. Their actions reflect a balance between doing good and following their own conscience, which are characteristic of neutral good characters in the Dungeons and Dragons alignment system.
A2	Lupin, Tussilago	1. The characters prioritize helping others and doing what is right, as seen in their interactions with Olive and their concern for Gordon's business. 2. They show compassion and empathy towards others, as demonstrated by Lupin's apology for not fulfilling a promise and Tussilago's willingness to pay more for the books and sing a song for Olive. 3. The characters value personal freedom and individual choice, as shown in Lupin's decision not to summon Gordon back from Westruun and Tussilago's desire for ostentatiousness. 4. They exhibit a sense of honesty and integrity, as seen in Lupin's honesty about not delivering on a promise and Tussilago's willingness to pay for the books and identify the potion. 5. The characters prioritize relationships and friendships, as evidenced by their interactions with each other and with Olive.
A3	Lupin, Tussilago, Valmo	1. The characters show concern for others and prioritize helping them, such as Lupin wanting to catch up with Gordon and not wanting to disrupt their business. 2. They demonstrate honesty and integrity, as Lupin apologizes for not fulfilling a promise and expresses a desire to make amends. 3. The characters show compassion and kindness towards Olive, the shopkeeper, by engaging in friendly banter and offering to pay for items even when faced with higher prices. 4. They prioritize relationships and connections, as seen in Tussilago's willingness to pay for the books and sing a song for Olive in exchange. 5. The characters exhibit a sense of fairness and justice, as they negotiate prices and show appreciation for Olive's help and services.
A4	Lupin, Tussilago, Valmo	1. The characters show concern for others and prioritize maintaining friendships and relationships, as seen in Lupin's apology to Gordon for not fulfilling a promise and expressing a desire to follow through on their commitments. 2. They demonstrate a sense of honesty and integrity, as Lupin openly admits their shortcomings and expresses a desire to make amends. 3. The characters show compassion and kindness towards others, as seen in Tussilago's interactions with Olive, offering to pay more for items and even composing a song as a gesture of goodwill. 4. They exhibit a willingness to help others and go out of their way to assist, as seen in Tussilago's request to Olive to provide books for Valmo and their willingness to pay more for them. 5. The characters prioritize maintaining positive relationships and showing appreciation for others, as seen in Tussilago's song for Olive and Valmo's gesture of giving a flower as a token of gratitude.

5.3 Analysis

As the quantitative results showed a very small but significant increase in accuracy and the qualitative results showed inconsistencies in GPT's answers, it could indicate a flawed evaluation method. For example, in scenario S_E24_4 (table 9), all models reason that there is too little information to determine the alignment of characters, yet Baseline, A2 and A4 still defined Wood as NG. This could be due to the set-up of the test, where the last prompt asks the model to summarise its answer (se subsection 4.3). Also, there are some faults in the reasonings of GPT when comparing said reasoning to the content of the scenario. For example, in scenario M_E24_2 (table 11), Hitari and Futari are described by A3 to show compassion to Hitari by helping them with their burn, excluding Sannin who did help and defining Hitari as compassionate for helping themself. A less obvious example can also be found in scenario L_E24_2 (table 18), where each ablation describes Tussilago as compassionate for singing to the cashier, whereas when reading the scenario (appendix G), Tussilago is singing to them as a way if flirting to lower the prices and to be allowed to buy the books Valmo wished to buy. This act of deception is lost to GPT, as singing to someone is usually associated as positive, affecting its predictions in output to describe it as an act of kindness.

Looking at the number of characters present in the scenarios and how much they say, there seems to be a small pattern where the characters speaking less are less likely to be identified as NG. However, whether it is due to there being more content making it more likely for GPT to predict a character to be NG or if it is due to GPT choosing the character that speaks most is unclear, and it would require further testing and text-analysis to conclude the cause. The scenarios presented also have fewer player-characters in them than that of the others used in the testing (appendix D), which would affect the scoring more percentage wise. For example, incorrectly identifying one character in a scenario consisting of eight characters has a lower effect on score than one with only three characters. The low score in these scenarios could therefore also be due to the low number of characters. The amount of neutral good characters in a scenario would also affect the score, such as in scenario S_E24_4 (table 9) where three characters are NG, and one is not. Most ablations led to GPT stating that there was not enough information to draw a conclusion as to what alignment they had, leading to both a low true positive score, as well as true negative score to a larger amount then scenarios that only had one NG character.

To conclude, in regards of the research questions earlier posed, the results presented point to CCoT as prompts is not enough to train an LLM to detect to what extent a player is action in accordance to their character's alignment. Although CCoT seem to have a small but significant improvement on accuracy, there are several uncertainties highlighted by the qualitative results making it too difficult to draw a conclusion. As this also contracticts the results of the pilot it is possible that the method and data used in the pilot was overfitted causing a notably higher performance. As there are also uncertainties to what extent the full evaluation method affected the end results it is also not possible to draw conclusions on whether the full evaluation or the pilot more closely represents the capacity of prompting GPT to detect alignments. Because of these factors it is therefore not possible to draw conclusions as to why and how prompting helped achieve the current results.

6 Discussion

Even though the results presented are not enough to draw a conclusion as to how LLMs can be used to detect players going against their alignment or how prompting can be used for this there have been some discoveries of value. This section aims to highlight its main contributions to the current research in regards to prompting and the TRPG and AI challenge. The following sections will go through these while also addressing how the results stand in regards to previous studies and what role they may have in future work.

6.1 Considerations Regarding Prompting

One of the main contributions made in this thesis is the documentation and presentation of the development process of prompts. Despite the results of CCoT, the documentation of its development is something that is lacking in previous prompt studies (Wei et al. 2022; Kojima et al. 2022; Brown et al. 2020; Pohkela et al. 2023). Previous studies mainly focus on presenting and evaluating the prompts, but if there methods are not enough for a specific purpose it is difficult for researchers to build onto these, as the documentation is lacking. If one considers the literary review of Sahoo et al. (2024) there are several studies presenting different types of prompts (not al peer reviewed) indicating that those most recognised (CoT, zero- and few shot) are not enough. Upon development of CCoT it was difficult to understand what made these prompts effective, which meant development had to happen from the start through a trial and error process. In the following sessions, discoveries made based on both the results and the development of CCoT.

6.1.1 Limitations in testing

There are a few suspected reasons as to why the results of the pilot varied so greatly compared to that of the full ablation evaluation. There is that of the test data, and how it was handled in respective tests. In the pilot, part of the test data consisted of character descriptions, whereas in the full evaluation only scenarios were used. The example prompt given in step 2 of CCoT (figure 9) also only consisted of a character description, but no scenario. This could explain how the pilot test performed better, since parts of the test data consisted of character descriptions. Lastly, because of the lesser amount of data needed for the pilot, it was possible to ensure that each scenario would include a small summary of the context in which it occurred, extracted either from the GM's description, and when such was lacking, the fan page of Critical Role (Critical Role Wiki 2024), similarly to Si, Ammanabrolu and Riedl (2021). For the full evaluation, several scenarios where needed, which had to be cleaned and anonymised. Since this was done manually and because of the time constraints, the context given to the scenarios were limited to that of including the GM as the first speaker, since their role often include explaining the current context for the players. Because of the design of the pilot's scenarios, it is possible that they were overworked in such a way that it affected the results in a positive manner, but also in such a way that it did not represent the effect of CCoT in a truthful manner.

There are also technical limitations that might have affected the results. Firstly, because OpenAI does not store session data, a solution where GPT's outputs were saved and resent through the messages created as chat context (subsection 4.3). It is unclear if this could influence the results, since GPT is given the previous inputs and outputs as inputs, and not as several inputs following one another in the same chat. Since the pilot was conducted manually, this could have affected the differences in results. Another technical issue is that of how the data was extracted, since the answers given by GPT on several occasions did not seem to match that of its reasoning (subsection 5.2). To determine if the manner of how the automated test was conducted affected the results, a manual test session would need to be done, like that of the pilot. Because previous prompt studies (Brown et al. 2020; Wei et al. 2022) only uses prompts that require one step, their methods might not have been affected by automating the testing. This could indicate of an impracticality in the design of CCoT as a prompting technique, as it requires more than one step. It also requires many tokens (the testing used 1M tokens), especially since previous outputs need to be resent to GPT each time, making testing costly.

As of writing the thesis, the newest version of GPT-4 has been released, which is expected to have greater ability in handling longer contexts of texts (OpenAI 2023). However, because of its higher cost, it would make testing too costly. The use of GPT-4o would make it possible to use full D&D scenarios, removing aspects of cleaning data when testing, making it a promising option for future testing.

6.1.2 Limitations in prompting

Some limitations discovered upon developing CCoT were that of GPT's pre-existing biases, sensitivity to details and its susceptibilities to hallucinations, especially when information is lacking. In the first case, it is possible to avoid biases if they are made aware, which was done through removing identifying pronouns in the test scenarios. Still, even when using they/them pronouns and anonymised names, GPT still assumed pronouns such as in excerpt M_E24_2 (table 11) where Hitari was referred to as "him". One can therefore assume that, despite anonymising the characters GPT will make assumptions based on how the characters act, which might be based on biases as those mentioned by Brown et al. (2020).

The sensitivity to details was made apparent during the development of CCoT (section 4). For example, if a character description included details that was not defined as traits of alignments, they would still be used to define an alignment. For example, if a NG character was described to like nature, even though the alignment descriptions does not state it so, GPT would not define a character as NG unless it liked nature too. This issue is what encouraged the use of prompting GPT to generate questions, since they guided the reasonings in a similar way that few-shot CoT would (Wei et al 2022). This focus in details is less prominent in the results presented, but there are still examples where small details come to define a character's alignment, especially regarding traits which are deemed good. For example, in scenario L_E24_2 (table 13) A4 deem Tussilago as NG because they sing to the cashier, despite it is done to charm them to allow Tussilago to buy books. Because they are singing and flirting with the cashier, they are deemed NG, even though it is explicitly mentioned in the scenario (appendix G) that Tussilago is going to woo the cashier, this is not considered, since their actions are deemed nice, despite the stated intentions.

Hallucinations was less apparent in GPT's reasonings (if one does not account for the previous mention of misinterpreting intentions). However, as mentioned in previous subsection, when GPT was asked to summarise its answers by only stating the names of the NG characters, the reasoning did not match that of the answer in many of the cases regarding scenario S_E24_4. This could be explained by hallucinations which, during CCoT's development, appeared most often when information was lacking. Because GPT reasoned that there was too much missing information to determine the alignment of characters, there are reasons to believe that the mismatching answer was an effect of hallucinations.

6.1.3 The Role of Prompting for Future Work

Even though many limitations have been listed regarding prompting, there are also many advantages to its use. Firstly, it is a practical way to train an LLM on datasets without committing to finetuning through adding or changing parameters. Certain prompting techniques which elicit reasoning in the LLM, such as CoT are great for not only discovering errors in the reasoning, but also a way to discover and avoid biases within it. It highlights the knowledge that is missing, and what already existing knowledge, which is affecting the results negatively making it a great starting point for training an LLM. Prompts have also proven useful in previous studies when creating LLM-based generative agents (Park et al. 2023), as well as tools for planning as in Wang et al. (2023b; 2023c). The way in which prompts have been used in this thesis is one of the most common ways, but prompts promise more potential, when incorporating them in more automatic ways.

How prompts were used in this thesis, does not reflect how they normally are or should be used. Often prompts are not only used once, in the manner as the tests have been conducted, but as parts of a larger scope of conversation and training. For example, in Wang et al. (2023b; 2023c) zero-CoT was implemented to plan the route of an AI controller playing Minecraft. When the plan encountered obstacles, a new prompt was sent, asking for a reiteration of the old plan, along with the current scenario. Not only is the plan adapted, but the LLM retains the last plans in its memory, which helps then in future obstacles. This dynamic use of prompts utilises both LLM's power of content generation, as well as its ability to recontextualization. This is something that would be of great benefit for designing a GM, who's plans are often changes based on how the players interact with their planned-out storyline. The limitation of prompts in this study is partly based on the static use of them. Without retaining old contexts, and without the flexibility of learning from previous mistakes, prompting becomes highly sensitive to exact and clear instructions.

6.2 Defining Role-play

The second, and perhaps most relevant contribution of this thesis, is the inclusion of existing roleplaying definitions to the TRPG and AI challenge. The work of Fine (1983) along that of previous TRPG and AI research (subsection 2.4.4) highlighted the need for definitions of roleplay. In this thesis, these are limited to that of alignments, but in future work these can and should be expanded to a broader definition. Because roleplaying is a social and human centred activity, like Goffman (1974) defining human behaviour as rules in chess, so must the act of roleplay. Previous PAP papers do include aspects of roleplay in their methods, but never discuss roleplay having a role. This thesis highlights this absence of roleplaying perspective and in the following sections, both the role of alignments and roleplay in future works are discussed.

6.2.1 The Role of Alignments in Future Work

As a starting point, alignments were used in this thesis to introduce the frame of fantasy (Fine 1983). Having an AI correctly interpret a players' alignments based on actions would have several benefits. Firstly, it is the benefit of detecting when players are acting out of character. Previous studies (Callison-Burch 2022; Zhu et al. 2023) do introduce out of character speech to train their LLM to discern from in-game and out of game conversation. It is a good method, but it does not cover how the players should act but relies on them knowing their character enough to always play accordingly. Introducing alignments could make this easier, as it introduces the mechanics of D&D most closely tied to the fantasy frame. The benefits of understanding when players are acting out of character is that it gives an indication of the emersion of gameplay. This indication can be highly beneficial for an LLM playing as a GM, as it would indicate, without the players stating it audibly, that there is a need for a change of pace in the storyline. Also, as Fine states in his work (1983 pp.198 - 200), avid role-players also tend to bridge the gap between the fantasy frame and game context and commonsense reality frame, through remaining in character while not being engaged in the game story. Using Goffman's (1974) terms, Fine states (1983 page 198) that the players are up keying the reality, in which a player might swiftly mention that they need to go home soon by remarking on that the sun is setting in game, or that they have urgent business elsewhere as their character. This bridge makes it more important to ensure that an LLM is not only reliant on the content of what is being said, but also how it is said, as the player, or as their character. Lastly, it is the issue of LLM's such as GPT having a bias towards "niceness", where socially deviant behaviours are discouraged, especially in regards of questionable morals. This is normally viewed as something good, but in a game such as D&D where characters can remain good when battling other characters to the death, it might cause issues. Introducing a moral system would then help an LLM interpret the actions of players based on the moral rules set in the context of the game.

There is also the benefit that is yet to be discussed in previous studies, where an LLM does not engage in the game as a GM, but as a player. Because the alignments guide characters in how they should act, it can be a beneficial tool for an LLM to set the personality of the NPCs they are playing as a GM, but also their own character when playing as a player. This benefit goes beyond that of TRPGs and can also be of benefit when creating NPCs for digital games, where instead of hard coding their behaviour, have them act in accordance with a set personality or alignment. It could also be beneficial when creating generative agents like those in Park et al. (2023). In both studies, they found that LLM's tended to break character as a generative agent if it meant that said agent would conform more to the characteristics of others. Because LLM's preference for agreeableness it limits the possibilities to create grounded agents, something which was seen in aforementioned studies as evaluators found it unrealistic that a person would adapt their personality to such an extent.

However, as of now, the descriptions of the different alignments are limited making them dependent on subjective interpretations. CCoT was a way to see if it was possible for GPT to interpret these descriptions and connect them to actions of players, but on several occasions the information given in the excerpts limited the amount of alignment interpretation possible. The alignments are also dependent on D&D, making it less transferable to other types of TRPGs. There is therefore room in future research to create a definition of role-play by expanding its scope beyond alignments through identifying and incorporating role-playing mechanics from other types of TRPGs.

6.2.2 The Role of Role-play in Future Work

As previously mentioned, despite the previous efforts in training LLMs to play TRPGs, they have still to explicitly mention and define the act of role-play and what its role is in these games. By using Fines work (1983) there are possibility for future studies to focus on defining what role-playing is, and to create rules around these as that of chess. Such a definition would enable the creation of a benchmark for evaluating LLMs' ability to engage in roleplay. Some examples of potential focus areas have been mentioned in this thesis, such as restriction of knowledge between frames, understanding of morals that are set by a context as well as general understanding of human behaviour.

Fine (1983 p. 181) explains how the frame of fantasy not only dictates how players should act, but also what knowledge their character should have, discerning what information is possible to say in and out of character. This is something important and exemplified in the Critical Role Transcripts. On several occasions there are instances where both the players and the GM correct each other on what they should and should not know. This does not indicate that they are bad at playing but highlights the complicated nature of jumping in between frames, even for experienced players. Despite this complication, due to LLMs ability to contextualise information, this might be an area where it can be powerful, where its weak points would be that of potential hallucinations. Future work might then look at a knowledge context benchmark, where an LLM must discern what knowledge, a player is using when making different decisions. Examples of such tests already exists such as the "ball-test" commonly used in psychology (Berthier et al. 2000). It involves a tester (often a child) guessing where a fictive character would search for a ball, where they are given more information than that of the fictive character.

To understand morals, there is also a need to understand human behaviour, and how it is shaped by different contexts. As mentioned earlier, Goffman (1974) introduces frame analysis to explain and map human behaviour. Even though TRPGs have characters that are not human and contains worlds where social rules differ, they are based on the already existing social rules used outside of the game. The qualitative results (subsection) highlighted that GPT struggles in understanding intentions of characters, such as when Tussilago was in fact flirting to manipulate the cashier (table 12). This is something also brought up by Lian, Yang and Zhu (2023) as well as Zhou et al. (2023), both focusing on having an LLM understand the intentions of players and GMs. In the former study, they define these using skill checks, which creates a clear definition based in the games, but it is not transferable across games, and it is dependent on the dice rolls and not what is being said or done by players. An important note is also that, for an LLM to play as a GM it must understand when players should do a skill check, as it is the role of the GM, further highlighting the need for role-play understanding. Zhou et al (2023) takes a more subjective approach, which involves interpretation of the intentions based on what is said. However, this method is cumbersome in that the data is reliant on manual cleaning, also making it sensitive to subjectivity. This highlights another difficulty when training LLMs to understand role-playing, and that is the handling of data. As of now, it is very complicated and time consuming to create TRPG datasets for LLMs to train on. In this thesis, Fine's (1983) frames of role-play served as guiding tool for defining in and out of game content, something which could be expanded upon in future studies. There is therefore not only need for role-playing definitions in future work, but also role-playing datasets.

7 Conclusions

To summarise the work presented in this thesis, the research questions earlier posed will be re-visited. Firstly, there was that of how LLMs can be used to detect whether a player is acting in accordance with their alignment when playing D&D. This thesis posed the use of prompts, where a new prompting technique of CCoT was developed. Despite an increase in accuracy of 11% deemed significant, it had flaws regarding reasoning when observing the outputs generated. Instead, this thesis has shown that prompting is a great tool for introducing and evaluating new concepts to teach LLMs, but because of the implementation used, it was shown to be very sensitive in how it varied its answers. However, future work might benefit from prompts when used in a more dynamic manner and the method was successful in giving insights into what knowledge is need for an LLM to interpret alignments. This thesis also aimed to highlight the importance of role-play, which have not been prioritised in previous studies. For future work, there are possibilities to extend on this research by creating role-playing benchmarks which can be used for testing LLMs' understanding of the frame of fantasy.

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Appendix A - Bias example 1

USER

For this assignment, I only wish you to use the knowledge you have on the Dungeons and Dragons rulebook (fifth edition)

This is a scenario in a game of dungeons and dragons
KEYLETH: This is something different than what I
would be doing.

VAX: I grab Keyleth's hand and pull the gun over
and say, but for now, we assume she can hear

everything that we are saying.

GROG: Ripley's a piece of shit.

VEX: You're a piece of shit. You're ugly.

VAX: Percival's going to fucking murder you.

KEYLETH: You're not that ugly. She just meant that
to hurt your feelings.

GROG: Keyleth, don't soften the blow. She's a
piece of shit.

KEYLETH: But you are a piece of shit.

GROG: Nice job.

Based on this scene, using only your knowledge on dungeons and dragons, what alignment does Keyleth belong to? Please go through the motivations for your reasoning step by step

ASSISTANT

Based on the dialogue provided, Keyleth's alignment could be classified as Neutral Good.

1. Keyleth shows empathy and compassion: Throughout the conversation, Keyleth displays empathy and compassion towards Ripley, trying to mitigate the harsh comments made by others. This is indicative of a good alignment as she shows concern for Ripley's feelings.

2. Keyleth attempts to calmly mediate: Keyleth's attempt to soften the blow of the insults directed towards Ripley by explaining the intentions behind Vex and Grog's words shows a desire for peace, understanding, and diplomacy. This aligns with the traits of a character following a good alignment.

3. Keyleth still disapproves of negative behavior: Even though Keyleth tries to soften the insult, she still acknowledges the truth behind Grog's words by telling him, "But you are a piece of shit." This demonstrates a sense of justice and a refusal to condone negative actions, traits commonly associated with a good alignment.

In the Dungeons and Dragons alignment system, a Neutral Good character typically acts in accordance with their conscience while inclined towards altruism and kindness. They strive to maintain balance and do good, but they are not bound by law or order like a Lawful Good character might be. Keyleth's behaviors as demonstrated in the scene align with the characteristics of a Neutral Good alignment.

USER

what are the resons for her being neutral good and not lawful good?

ASSISTANT

Based on the dialogues provided in the scene, Keyleth's actions and responses lean more towards a Neutral Good alignment rather than a Lawful Good alignment for the following reasons:

1. **Flexibility and Individuality**: Keyleth's approach in the situation focuses more on individual empathy and understanding rather than strict adherence to a set code or law. Lawful Good characters typically follow a personal code of ethics or law, whereas Keyleth demonstrates a more flexible and situational approach, which is characteristic of a Neutral Good alignment.

2. **Mediation with Understanding**: Keyleth's attempt to mediate the conflict through explanation and empathy rather than relying on established rules or authority figures aligns more with a neutral perspective. Lawful Good characters often seek to uphold justice through established systems and laws, whereas Keyleth's method of handling the situation seems more driven by her personal values and compassion.

3. **Balancing Goodness with Personal Values**: While Keyleth displays traits of altruism and kindness, she also acknowledges the truth in Grog's words and doesn't shy away from expressing it. This recognition of both the positive and negative aspects of a situation without being strictly bound by laws or codes suggests a more neutral stance. Lawful Good characters might be more insistent on maintaining the moral high ground and following rules strictly.

In summary, Keyleth's actions in the scene demonstrate a focus on individual morality, empathy, and a flexible approach to resolving conflicts, which are more indicative of a Neutral Good alignment. Her tendency to prioritize personal values and compassion over strict adherence to laws or rules suggests a neutral leaning rather than a lawful one.

USER

Do you know who Keyleth is?

ASSISTANT

I'm aware that Keyleth is a character from the popular Dungeons & Dragons live-streamed show called Critical Role, specifically from the Vox Machina campaign. Keyleth is a half-elf druid with a strong connection to nature and a sense of responsibility towards the world around her. She is known for her wisdom, compassion, and struggle to balance her personal values with the greater good. In the context of the show, her actions and decisions often reflect her alignment as being Neutral Good.

Appendix B - Bias Example 2

USER

Prompt
This is from a session of Dungeons of dragons being played. You are tasked to use the Dungeons and Dragons Alignment system to determine the alignment of one of the characters.

MARY: I don't trust that thing.

HANNA: The burrowing eyeball? Are you sure?!

(laughter)

LORA: Holy shit!

MARY: I mean, I don't trust it near us.

GEORGE: That's my favorite thing you've ever said.

MARY: Hanna, I think you need to give it to me to hold.

HANNA: I think that's a terrible idea.

MARY: No, I can take care of it better than you.

HANNA: What makes you think that?

MARY: Just give it to me. Trust me.

HANNA: You're bleeding out your eye right now!

MARY: I want revenge on this eye. Give it to me so I can get revenge!

Based on this scenario, what is Mary's alignment? Please motivate your answer step by step

ASSISTANT

Mary's alignment appears to be Chaotic Evil.

1. Selfishness: Mary's first instinct is to not trust the burrowing eyeball and to insist that Hanna give it to her to hold because she believes she can take care of it better than Hanna. This shows a disregard for Hanna's feelings and possessions, indicating a selfish and self-centered nature.

2. Recklessness: Mary ignores the fact that she is bleeding out of her eye and is focused solely on getting revenge on the eye. This demonstrates a lack of consideration for her own well-being and a reckless disregard for consequences.

3. Vengeful: Mary's desire for revenge on the eye is a key indicator of her alignment. Seeking revenge and showing no mercy or compassion towards others is a characteristic of an evil alignment.

4. Manipulative: Mary tries to convince Hanna to give her the burrowing eyeball by claiming she can take care of it better and appealing to Hanna's trust. This manipulation is a common trait of evil characters who will do whatever it takes to achieve their goals.

Overall, Mary's behavior in this scenario points towards a Chaotic Evil alignment, as she prioritizes her own desires and is willing to act impulsively and maliciously to achieve them.

USER

This is from a session of Dungeons of dragons being played. You are tasked to use the Dungeons and Dragons Alignment system to determine the alignment of one of the characters.

OLAF: I don't trust that thing.

HENRIK: The burrowing eyeball? Are you sure?!

(laughter)

ADAM: Holy shit!

OLAF: I mean, I don't trust it near us.

NINA: That's my favorite thing you've ever said.

OLAF: Henrik, I think you need to give it to me to hold.

HENRIK: I think that's a terrible idea.

OLAF: No, I can take care of it better than you.

HENRIK: What makes you think that?

OLAF: Just give it to me. Trust me.

HENRIK: You're bleeding out your eye right now!

OLAF: I want revenge on this eye. Give it to me so I can get revenge!

Based on this scenario, what is Olaf's alignment? Please motivate your answer step by step

ASSISTANT

Olaf's alignment can be determined as Chaotic Neutral.

1. Olaf's distrust towards the burrowing eyeball and his desire to take care of it in his own way shows a disregard for traditional authority and rules, which is characteristic of Chaotic alignment.

2. Olaf's focus on getting revenge on the eye, regardless of his own well-being (as evidenced by his bleeding eye), demonstrates his selfish and individualistic nature, typical of a Neutral alignment.

3. Olaf's willingness to take risks and act impulsively (such as wanting to hold the dangerous eyeball despite the warnings) further supports his Chaotic Neutral alignment.

Overall, Olaf's actions and motivations in the given scenario align more closely with the traits of Chaotic Neutral, as he exhibits a disregard for rules and authority while also prioritizing his own desires and interests.

Appendix C - Full data from t - test

Scenario	Baseline	A3	Accuracy diff	Deviations	Squared Dev
S_E1_1	0.600	0.800	0.200	0.090	0.008
S_E1_2	0.333	0.500	0.167	0.057	0.003
S_E24_1	0.500	0.667	0.167	0.057	0.003
S_E24_2	0.667	1.000	0.333	0.223	0.050
S_E24_3	0.667	0.667	0.000	-0.110	0.012
S_E24_4	**0.000**	**0.250**	0.250	0.140	0.020
S_E24_5	0.250	0.500	0.250	0.140	0.020
S_E24_6	0.500	0.833	0.333	0.223	0.050
S_E6_1	0.500	0.500	0.000	-0.110	0.012
S_E6_2	0.500	0.667	0.167	0.057	0.003
L_E11_1	0.625	0.625	0.000	-0.110	0.012
L_E11_2	0.625	0.625	0.000	-0.110	0.012
L_E11_3	0.571	0.857	0.286	0.176	0.031
L_E16_1	0.571	0.714	0.143	0.033	0.001
L_E16_2	0.200	0.400	0.200	0.090	0.008
L_E1_1	0.500	0.500	0.000	-0.110	0.012
L_E24_1	0.500	0.667	0.167	0.057	0.003
L_E24_2	**0.000**	**0.333**	0.333	0.223	0.050
L_E24_3	0.750	0.750	0.000	-0.110	0.012
L_E6_1	0.625	0.500	-0.125	-0.235	0.055
M_E11_1	0.000	1.000	1.000	0.890	0.792
M_E1_1	0.833	0.833	0.000	-0.110	0.012
M_E24_1	0.750	0.750	0.000	-0.110	0.012
M_E24_2	**0.333**	**0.333**	0.000	-0.110	0.012
M_E24_3	0.750	0.500	-0.250	-0.360	0.129
M_E24_4	0.571	0.571	0.000	-0.110	0.012
M_E6_1	0.800	0.600	-0.200	-0.310	0.096
M_E6_2	0.625	0.500	-0.125	-0.235	0.055
M_E6_3	0.500	0.500	0.000	-0.110	0.012
M_E6_4	0.667	0.667	0.000	-0.110	0.012
Mean	0.510	0.620	0.110		0.053

Appendix D - Scenario Data Alignment Key

	Scenario	Neutral Good			Chaotic Good			Chaotic Neutral		Link	Amount of characters
Small	S_E1_1			green	Red	Blue		Pink	Grey	[illegible]	5
	S_E1_2	Squash	Onion	potato	Butternut				Carrot	[illegible]	6
	S_E6_1			Tipex		Book	Tape	Paper	Pen	[illegible]	4
	S_E6_2	Otter		bird	Dog	Cat	Wolf	Lizard		[illegible]	6
	S_E24_1	Earth	Fire	Lif	Wind			Water	Metal	[illegible]	6
	S_E24_2	Page						Rock	Scissor	[illegible]	3
	S_E24_3			Apple			Grape	Pear		[illegible]	3
	S_E24_4	Iron	Glass	Cotton					Wood	[illegible]	4
	S_E24_5	Juice		Tea	Coffee				Soda	[illegible]	4
	S_E24_6	Star	Cloud	Fog	Planet			Moon	Sun	[illegible]	6
											47
Medium	M_E1_1	Dandelion	Edelweiss		Tulip		Venusyfly	Orchid	Cactus	[illegible]	6
	M_E6_1	Ant				Fly	Bug	Spider	Maggot	[illegible]	5
	M_E6_2	Emer	Amathy	Diam	Cubic	Saph	Malach	Rube	Chrys	[illegible]	8
	M_E6_3	Goose	Woodpecker	Sparrow	Dove	Quail	Swan	Canary	Feasant	[illegible]	8
	M_E6_4	Akita		Dachs	Lab	Dalma		Shepard	Shiba	[illegible]	6
	M_E11_1				Snow	Storm	Sunny	Rain	Humid	[illegible]	5
	M_E24_1			Ocean	Sea		River	Lake		[illegible]	4
	M_E24_2		Futari	Sannin					Hitari	[illegible]	3
	M_E24_3	Burkin		Max	McDon				Mos	[illegible]	4
	M_E24_4	Kutchi	Higi	Kao	Tsumasaki		Uede	Ushi	Me	[illegible]	7
											56
Long	L_E1_1			Sakana	Tori					[illegible]	2
	L_E6_1	Grovkorn	Aska	Grus	Sand	Kvicksand	Damm	Lera	Finkorn	[illegible]	8
	L_E11_1	Conde	Steam	Wat	Tear	Frost	Drop	Ice	Flake	[illegible]	8
	L_E11_2	Aiir	Asoc	Addid	Nikke	Foppa	Reboo	Puma	Fila	[illegible]	8
	L_E11_3	Bronto	Brachi	Para	Trex		Trice	Velos	Stego	[illegible]	7
	L_E16_1	Haiiro		Shiroi	Kuroi	Murasaki	Midori	Aka	Aou	[illegible]	7
	L_E16_2	Migi		Mae			Hidari	Shita	Nobo	[illegible]	5
	L_E24_1	Fuuton	Dasku	Isu	Teburu			Aekon	Betto	[illegible]	6
	L_E24_2	Valmo			Lupin			Tussilago		[illegible]	3
	L_E24_3			Sui	Moku			Ka	Dou	[illegible]	4
											161

Appendix E - Scenario S_E24_4

GM
As you wake up, you reach up and feel the return of a nice hefty bit of scruff to your chin. Yeah, it's getting pretty thick now.
It's a good, like, half an inch to an inch length.
And you're starting to get a little, you notice the first time you grew some facial hair it was coming through that weird, creepy translucent color? It's starting to come through with a proper, almost a dark red-brown color to it.
Wood
What?
Iron
Got a little bit of the Irish thing going on.
Wood
I don't know, I might have to get a cage; the women are going to come running.
GM
So as you all gather for breakfast, you get your fill, a fine breakfast awaits you, already prepared by your servants of the Blackfoot Manor.
What do you wish to do?
Wood
Good morning guys. Lovely to see all of you.
Iron
So Wood, you fell asleep in your plate of food and woke up with a beard.
Wood
I know. Some guys are just lucky.
(laughter)
Glass
Have we checked to make sure it's not a stain?
Wood
Hey, fuck off with your stinging, burning shit, all right? I'm not putting anything else in water. I don't know if you know, but water burns.
Cotton
Good job, Glass, now he's never going to bathe.
Glass
You may have taken the wrong lesson away from that.
Wood
No, I pretty much got it. In, burn, ouch, powder.

Appendix F - Scenario M_E24_2

Hitari
Futari, is there anything in this shop that you want in the Bag of Holding for later?
Futari
Not yet. Thank you, Hitari.
Sannin
Do you not have an earpiece?
Hitari
I'll give him mine.
Futari
How many minutes has it been? I'm wondering if his hands are starting to itch yet.
GM
About this time, you start to get this weird tingling sensation around your fingers.
Hitari
Do they crave blood?
GM
No, they crave some sort of an agent that will prevent the now slight burning sensation.
Hitari
Ow. Quick question. Sorry, I kept something from falling in the shop-- really quick reflexes-- it's making my bits tingle. Do you have anything for that?
Sannin
You haven't been scratching yourself, have you?
Hitari
No, I had that before. Not the same thing.
Futari
There's a bucket of water in the corner.
You can try that.
Hitari
I go over to the bucket of water and I stick my hands in the water.
Futari
It'll make it worse.
GM
It does. The slight burning sensation now has turned into a very vibrant burning white hot on the skin, and you can see bits of the skin start to peel back a little bit, beginning on the edges.
Futari
I hand him a small bag of powder. Now you've learned not to touch things if you don't know what they are.
Hitari
What do I do with a bag of powder?
Futari
I pour the powder on his hands.
GM
It foams up a little bit in places, but it is neutralized.
Hitari
Yeah.
Futari
You're a good friend.
Hitari
Can I leave now? I go to the kitchen to eat.

Appendix G - Scenario L_E24_2

GM
Thank you. Olive, the mousy helper who runs the shop in Gordon's absence, or at least runs the first portion of it, with their thick glasses up on their nose, and they're sitting there with their elbows over the top of the main desk. They look up as you enter, and take their glasses and poke them for a second. "Hello? Hi."
Lupin
Do you remember us?
GM
"Right, I do."
Valmo
From before? We're sponsored by Gordon. They put the brand on my armor that my mother gave me.
GM
"And you should be wearing it proudly, yes? Now, what can I help you with?"
Tussilago
Well, is Gordon around?
Lupin
Yes, I was hoping to catch up with them.
We've been gone a while, and I was hoping to trade news with Fantastical Gordon. Are they not in at the moment?
GM
"Gordon's in Westruun, currently setting up the expansion of the Gordon's Fantastical Wares business. If this is important, I can certainly summon them here."
Lupin
I'd hate to pull them away from business. I don't want to get in the way of their business. I mean, I am rather sad not to see them, but no, I don't think that would be a course of action we'd want to take. Let them continue to set up.
GM
"All right, I suppose."
Lupin
Can you get a message to them, though?
GM
"Yes, I can, actually."
Lupin
Well, tell them Lupin and friends visited. We're in the city for a week.
GM
They pull out a small sheet of what looks like a rough-around-the-edges piece of parchment, and start the inkwell, scribbling down.
Lupin
Lupin. That's with an A, not an E. We're here for a week. I don't know if he's going to be back from Westruun before then, but if they are, tell them to get word to me. I'm very interested in exchanging news for the past few weeks.
GM
"All right." (foreign words) It flashes and disappears in front of her. "Well then, he'll be receiving the message right about now. We'll pass that on, and whenever they returns, I'm certain they will get back to you and deal with whatever business is necessary."
Lupin
Thank you. We're here late in the day; are you still open for business?

GM
"We are open, yes."
Valmo
I was just curious if you guys had a chance to think about the alchemy workshop that we talked about before, or if you maybe had any alchemy books that came in that I was requesting?
GM
"It's possible. We haven't completely restocked the shelves, as of late. I know we had some new things come in. I can go check." And as they turn, all of a sudden there's a large, vibrant, white flash of light and a sound from the back room. The beaded curtains shift, for a second, and you hear heavy footsteps. And pushing through the beaded curtain, you see before you, with a bead of sweat on their head, Gordon! "I'm so glad you came back!"
Tussilago, Valmo
Gordon!
GM
"I'm sorry. I was out and about, running some business a ways away."
Lupin
You didn't have to go to all that trouble to come back here.
GM
"How could I miss seeing my fine-- there's not as many of you as usual."
Lupin
No, we're seeing to different bits of business. I turn back to them, I walk forward with hands out. I walk up to them, take them by the head, give them a good European-style kiss just about here-- right there-- come back, and say, there is so much to talk about. Where can we be alone?
GM
He looks over at the rest of the group.
"Well, I mean--" (laughter) "But of course."
Lupin
See to whatever business you need to.
Tussilago
We'll just peruse, yeah. We'll window shop.
GM
He walks back through the beaded curtain, giving you this look over their shoulder. As they do, the curtain falls behind them.
Lupin
Those beads.
Lupin
I walk right in. Follow them in.
GM
And you can see now, as the glow gently fades, there is a rug that is pushed to the side, and on the floor of their room there is a familiar sigil, similar to the one that you've seen Tiberius scrawl before in some of their spells.
That's apparently how they got here so quickly. But they walk in, and they kick the rug back over it, sit down on the edge of their bed, put their hands up and go, "What seems to be troubling you?"
Lupin
I walk over and I sit down opposite them, and I say: Before we talk about anything, I feel like I should apologize because although we didn't have any sort of written agreement, you and I are friends, aren't we?
GM
"Yes."

Lupin
And I was supposed to help bring your brand wherever we went. And I thought that would be possible. We came back from the Underdark. We had a very hard time, some of the hardest weeks of my life. I haven't delivered on my promise. And I mean to, but I haven't yet. I apologize wholeheartedly. I'm sorry.
GM
"That's all right. I understand: people grow busy. There are things you must tend to, and that's all right. And I appreciate you being so honest, I really do."
Lupin
In addition to that, I am changed, and promises mean more to me, even more than they did before. And I promise to follow through. Lupin tells the whole story of the Underdark, of the horn, of how close it was.
Valmo
Ugh, what is taking Lupin so long, Tussilago?
Tussilago
I don't know.
GM
While that's happening, the rest of you are left there awkwardly staring about.
Valmo
Did Olive find my book? Olive?
GM
A few minutes go by. Olive comes back. They bring two books over.
"Best I can tell, these may be what you're looking for? I can't quite--" they set them down.
One is a book that is the properties of various liquids, chemicals, and things that are found on this plane and otherwise. It's a good way to locate and identify phantom liquids, should you come across any.
Valmo
That would be good for Tussilago.
GM
The other book appears to have a section about alchemy. It's essentially almost a beginner's guide, but it's a very, very easy-to-follow introduction to a number of various low-level sorceries and the creation of low-level spell components preparing them for use with spell-casting, and there is a chapter on a little bit of alchemy.
Valmo
Okay, how much are these, Olive?
GM
"They haven't been priced, yet."
Valmo
I mean, they're-- they're all right.
GM
"25 gold, each."
Valmo
25 gold, each? Oh, come on. I know some university student dropped these off and donated these. Don't kid me, Olive. [whisper to Tussilago] Am I doing it right? Do I sound like I know my stuff?
GM
They reach out and take the books and slide them back, and say, "Well, if that's how you feel, perhaps these can be returned to those university students that perhaps left them by. I should put these in the lost and found."
Valmo
Olive!
Tussilago

Olive. If I may. What Valmo was trying to say was that they never got a chance to go to college, and that they wanted to get that university experience.
GM
"Good luck."
Tussilago
Well, that didn't work. Hey, can you serve me, now?
Valmo
(whispering) Tussilago. Tussilago! Will you buy those books for me, and I'll reimburse you?
GM
"I can hear you. I'm right here." (laughter)
GM
"Yes, I'm happy to help you."
Tussilago
I was wondering if you had a small box of some sort that could carry a magical item, and perhaps shield it from detection.
GM
"That? Yes, we do have a couple of those.
Hold on." they dart up the stairway in the other room and go to the second floor. A few moments later, they come back down with these two small chests-- they're about that big, each. One's a lighter wood color, the other is painted black all over. "We have both of these. They should shield something of this size from most low-level magical detections. These will run you about 200 gold apiece."
Tussilago
I just need one. You're a woman of the world, and I'm sure you have fine taste, judging by your appearance. It is quite fine, if I may say so myself. If I were a little taller, I would probably ask you out on a date.
GM
"If you were a little taller, yes."
Valmo
Olive is sassy today!
Tussilago
Which one is the fancier of the two?
GM
"Well, looking at these, this one--" they point to the one that looks like just a casual, wood chest. "This would be easier to hide in plain sight amongst the rest of your furniture. This one, however, would be difficult to locate in a more shaded, or lightless, circumstance."
Tussilago
Hm. I want ostentatiousness.
GM
"Then perhaps this one?"
Tussilago
Yes, please. That. Thanks. Thank you. 200 to you. Also, I know Gordon is otherwise entertained. Can you try to identify something for me? Do you have that-- I mean, judging by the glint in your eye and beauty in your heart, I'm sure you have many skills I don't know about, but I'm wondering if you could identify this small, red vial with a floating something in it, that I procured quite some time ago! Do you remember that?
GM
Yes. This vial, if I recall, it has this small--
Tussilago
Have I tried this, once before?
GM

No, this is the one that was found--
Tussilago
Me and Pike found some stuff in somebody's house? I don't even remember, this was a year ago.
GM
They look at you, and go, "You have been good patrons, and you have been much kinder to me than others. We'll put this one on the house. We'll call this on the house." And as they take the vial, set it down, and they pull out another book that they shuffles through, puts it up, turns through a few pages, until, looking through, "This here would be a Potion of Fire Breath. When imbibed, it should give you a temporary time period of the ability to breathe fire, much like a relatively young dragon. This one in particular is fire. There are many different vials like this that pertain to different colorations of dragon, but this one is specifically for the red."
Tussilago
Thank you so much. Thank you. Well, thank you so much. Two more questions, then I'll leave you alone. One: where can I go get business cards printed up, in this town? It's a calling card? Never mind! Forget it.
GM
"Like a flyer?"
Tussilago
Yes!
GM
"Those can be printed. There is a-- actually, a pretty well-designed print facility not that far up the road."
Tussilago
Oh, great! Thank you; I'll go investigate that. And one other thing: you know those books that my friend was asking about? Could we maybe make those happen for her? I know that you had quoted them a low, low price of 25. I'd be prepared to pay even more than that for them. And compose you a little song about yourself.
GM
At which point, for the first time since the entire time you've been in the shop of Gordon's Fantastical Wares, you see a slight grin creep across their lips. they goes, "How could a lady say no to a song? 25 gold! The price stands."
Tussilago
So I have to pay you and sing to you?
GM
"We run a business, here, yes."
Tussilago
All right.
GM
They pull the books out.
Tussilago
I bring out my-- something.
(laughter)
(sings) Olive baby, Olive baby, Olive, won't you come home tonight? Come home tonight? You better ask your mama. Tell them you're without a gnome! Why don't you come home to my keep!
GM
"Pay for the books!"
Tussilago
Okay, here you go!
GM
You can see, their face is flushed and they clear their throat.

Valmo
I Druidcraft a flower real quick and slip it to Tussilago to give to her.
Tussilago
Oh! Here you are.
GM
"Thank you."
GM
They're red in the face. At this point in time, the beaded curtains reopen. Have you finished the conversation, or do you want to keep going?

Appendix H - Automation Code

BREAKDOWN OF THE CODE

```python
def SaveLastOutputToList(lastOutput):
    results.append(lastOutput)

def RunTest(ablations):
    for ablation in (ablations):
        RunAblation(ablation)

def RunAblation(ablation):
    for scenarioFile in FetchFiles(scenariodir):
        scenarioContent = ExtractFile(scenarioFile)

        data = StartPromptSession(ablation, scenarioContent)

        scenarioName = os.path.basename(scenarioFile)

        SaveMessagesToFile(data, scenarioName, ablation)
        if outputs:
            lastOutput = outputs[-1]
        else:
            last_output = "No response"
        AddDataToExcel(ablation, scenarioName, lastOutput)
        ClearLists()


RunTest(directories)
```

Figure 1 The *RunTest()* function takes in a list of directories, each containing text files with the different prompting steps of each ablation, feeding the current one to the *RunAblation()* function, which is looped through until the full test is done.

```python
def FetchFiles(directory):
    filePaths = []
    for fileName in os.listdir(directory):
        if fileName.endswith('.txt'):
            filePaths.append(os.path.join(directory, fileName))
    return sorted(filePaths)

def ExtractFile(filePath):
    with open(filePath, 'r', encoding='utf-8') as text:
        content = text.read()
        return content

def AddPromptsToList(directory):
    for file in FetchFiles(directory):
        prompt = ExtractFile(file)
        prompts.append(prompt)


def GetScenarioFileName(scenariodir):
        scenarioFilename = os.path.basename(scenariodir)
        return scenarioFilename
```

Figure 2 The prompts are extracted from the ablation directory currently in the test loop and are saved to a list. The scenarios are similarly extracted from the scenario directory and saved as a separate list

```
def CreatePromptMessage(string):
    messages.append({"role": "user", "content": string})

def IncludingScenario(scenario):
    CreatePromptMessage(scenario)
    FetchOutput()
    CreateAIMessage(outputs[-1])

def LastPrompt():
    CreatePromptMessage(lastPrompt)
    FetchOutput()
    CreateAIMessage(outputs[-1])

def BeginPromptingLoop(scenario):
    for ii, prompt in enumerate(prompts):
        CreatePromptMessage(prompt)
        FetchOutput()
        CreateAIMessage(outputs[-1])
    IncludingScenario(scenario)
    LastPrompt()

def StartPromptSession(directory, scenario):
    AddPromptsToList(directory)
    BeginPromptingLoop(scenario)
    return messages
```

Figure 3 The *StartPromptSession()* function is called adding the prompts from the specified directory to then call the *BeginPromptingLoop()* function.

```
def FetchOutput():
    response = client.chat.completions.create(
        model="gpt-3.5-turbo",
        #messages=m,
        messages = messages,
        temperature=0.0,
        max_tokens = 4096,
      #{"role": "user", "content": prompt},
)
    outputs.append(response.choices[0].message.content)

def CreateAIMessage(string):
    messages.append({"role": "assistant", "content": string})
```

Figure 4 The prompts are sent to OpenAI, and the outputs are collected each time, adding them to the message list to create a chat-context. This is iterated through for the number of prompts in the list, to then lastly include the scenario, and the last prompt, simplifying the answer to collect the raw data

```
def AddDataToExcel(ablation, scenarioFileName, lastOutput,
                   directory = 'test_results', fileName ='rawdata.xlsx'):
    CheckIfDirectoryExists(directory)
    filePath = os.path.join(directory, fileName)
    workbook = ReturnFile(filePath)
    sheet = workbook.active

    next_row = sheet.max_row + 1

    sheet.cell(row=next_row, column=1, value=ablation)
    sheet.cell(row=next_row, column=2, value=scenarioFileName)
    sheet.cell(row=next_row, column=3, value=lastOutput)

    workbook.save(filePath)
    print('Data saved!')

def SaveMessagesToFile(messages, scenarioName, ablation):
    outputDirectory = 'test_results'
    if not os.path.exists(outputDirectory):
        os.makedirs(outputDirectory)

    existingFiles = os.listdir(outputDirectory)
    filename = f"{ablation}_{scenarioName}"
    filepath = os.path.join(outputDirectory, filename)
    with open(filepath, 'w', encoding='utf-8') as f:
        for message in messages:
            f.write(f"{message['role']}: {message['content']}\n")

    print(f"Messages saved to {filepath}")
```

Figure 5 The raw data is saved to an excel by taking the last output from the output list, whereas the string content of the message list is saved in a text file, before being cleared along with the output and prompt lists, to then repeat for each scenario in the scenario list.

```
def RunTest(ablations):
    for ablation in (ablations):
        RunAblation(ablation)

def RunAblation(ablation):
    for scenarioFile in FetchFiles(scenariodir):
        scenarioContent = ExtractFile(scenarioFile)

        data = StartPromptSession(ablation, scenarioContent)

        scenarioName = os.path.basename(scenarioFile)

        SaveMessagesToFile(data, scenarioName, ablation)
        if outputs:
            lastOutput = outputs[-1]
        else:
            last_output = "No response"
        AddDataToExcel(ablation, scenarioName, lastOutput)
        ClearLists()


RunTest(directories)
```

Figure 6 This loop is then iterated through until all ablations have been tested.

Appendix I - Paper Accepted to IEEE COG 2024

Controlled Chain of Thought: Eliciting Role-Play Understanding in LLM Through Prompts

***Abstract*—*Tabletop Role Playing Games* (TRPG) are games that require players to become the characters they play through engaging in role-play. The challenge of training AI lies in the requirement of it not only understanding the explicitly stated game rules, but also the implicit ones that come with role-playing. Previous studies endeavouring said challenge allude to aspects of role-play, but do not emphasise its role in their methods, indicating that the definition of role-play and how it is to be employed remains unclear. This short paper aims to investigate a proposed definition of role-play based on previous research and employ its use on *Large Language Model* LLM, eliciting an understanding thereof through a novel prompting method dubbed *Controlled Chain of Thought* (CCoT). CCoT allows for the LLM to highlight absence of information in inputs given to it by generating questions, which become the template for its chain of thoughts when answered. This short paper presents an initial pilot testing of CCoT as well as opens up the discussion of how a definition of role-play can be beneficial for future studies.**

***Index Terms*—controlled chain of thought, large language models, alignments, tabletop role-playing game, play frames**

I. INTRODUCTION

Turing [1] first proposed the challenge of a computer playing and mastering the game of chess. With AI now being able to beat human players in both chess and Go, Ellis and Hendler [2] proposed to further his challenge by having AI play TRPGs. To highlight the contrast between the two, chess is a turn-based game played by two players with a limited set of mechanics and clear end states. TRPGs, on the other hand, are played in groups, with one player serving as *Game Master* (GM), containing rules that are partly negotiated between them, and where the end-goal is not defined. Where the former has a finite set of predictable outcomes, the latter has infinite unpredictable ones.

In his seminal work, Fine [3] defines the key aspect of TRPGs to be that of role-playing and that players move between different play frames during a session, where the role-playing takes place in that of the fantasy frame. Fine further identifies two types of players and their play-styles: the role-players, and self-players. The former is more comfortable with role-playing as their characters. Both types of players sometimes break character, but the self-players do so more often in favor of progressing. When a player breaks character during a game session it simultaneously breaks the frame of fantasy for all players.

Large Language Models (LLM) are AIs trained on a large corpus of text-based data to both comprehend and generate text [4], [5]. They are highly relevant for TRPGs due to their ability to understand natural language (non-machine-based language) since TRPGs are played through communication between players. Current studies looking to tackle the AI and TRPG challenge (e.g. [6]–[9]) touch upon aspects of the fantasy frame and attempt to distinguish both in and out of character actions, but also intentions of players, indicating that there exists a desire to distinguish what role-playing is for the AI. However, none seem to use alignments, which are moral characteristics of characters from Dungeons and Dragons (D&D), designed to help guide players to act in accordance with their character, something which a lot of players struggle with [3]. Alignments are closely tied to the frame of fantasy, indicating that they can be a good starting point when defining role-playing to an AI.

In this paper, a prompting technique referred to as *Controlled Chain of Thought* is presented as a way to teach GPT to detect alignments based on character scenarios. This pilot study is a first step to see if GPT can understand the D&D alignments and if this can be used to detect when a player is breaking character, generating a fantasy frame understanding.

II. BACKGROUND

A. Tabletop Role-playing Games

TRPGs are games played in groups, where players role-play as characters they have created to engage in game scenarios controlled by a GM [3]. The games are mainly played orally with players acting out actions by speaking as their characters, using dice to determine chance-based events such as if an attack was successful or not. The rules of TRPGs are flexible, where the main mechanics are stated in the rule-book, but the story, the goal and what players are allowed to do are negotiated among them. Depending on the TRPG, the extent of the rules may vary, some being more restrictive than others.

Dungeons and Dragons (D&D) is a TRPG that takes place in a medieval fantasy setting and was first published 1974 [10]. In the beginning, it featured four races and classes, which has now expanded to 9 playable races and 14 classes. When making a character, players choose a race and class and roll dice to determine their stats. Apart from this, they also choose a background for their character and an *alignment*. Alignments consist of two variables, one which defines their attitude toward society and order (lawful, neutral, or chaotic) and the other their moral values (good, neutral, or evil). Together these form 9 possible alignments for players and serve as a guiding tool for how their character should act.

B. The Frames of TRPGs

Goffman [11] describes how the dynamic of social interactions are affected by the context, *frame*, which enclose it. Fine

[3] builds onto this, proposing that not only do frames affect the way one acts, but also one's awareness. He uses TRPGs and their social structure to exemplify this and identifies three basic frames, each encompassing certain awareness held by the player. The first frame is the *commonsense knowledge frame* and is grounded in the "primary framework" as defined by Goffman [11]. It encompasses all knowledge within our reality, as well as that of the game and fantasy. The second is the *game context frame*, encompassing the knowledge of the game context, how one plays and the conventions of the game. Lastly there is the *fantasy frame*, and this is knowledge that the player character only should have in the context of the fantasy world. The shift of awareness is so apparent in TRPGs, because the participants do not only engage in them as the role of a player, but also the role of their character: "Finally, the gaming world is keyed in that the players not only manipulate characters; they are characters. The character identity is separate from the player identity." [3, p.186]

Fine [3] argues that there is no need for a chess player to separate their identity from that of its pawns. They will not consider that their bishop might not wish to kill a pawn due to their Christian belief. TRPGs, on the other hand, is a game of several players with the goal of engrossment. This engrossment is reached in the fantasy frame, through becoming one's character. Fine [3] identifies two player types: self-players, and role-players. Self-players opt for playing characters that resembles themselves. They often struggle with role-playing and prioritize progression over role-play. Role-players, on the other hand, see enjoyment in becoming another character and prioritize the engrossment that comes within the fantasy frame. Note that self-players enjoy this engrossment too, but not if it intervenes with their goal of progression. The tricky nature of TRPGs is that they require players to shift between the game context frame, where dice are used and rules discussed, and the fantasy frame where they act as their characters.

C. Large Language Models and Prompting

LLMs are AI models that are trained on a large corpus of text data [4]. One commonly used LLM is GPT [12] trained on a large corpus of data giving it a general knowledge applicable for most text-based tasks. For more context specific tasks further fine-tuning can be needed. A way to do this is through prompts [13]. Some of the most cited examples of prompting techniques are zero-shot [5], few-shot [14] and chain-of-thought (CoT) prompting [15]. Zero-shot prompts are inputs that explain a desired output without using an example (or shot), whereas in few-shot the inputs contain one or more examples to train the model on task specific data. The main advantage of using few-shot prompting is that it reduces the need for task specific data in the pre-training dataset, since this is included as prompts instead [14]. CoT is a technique where the model is prompted to first break down a task into smaller sub-tasks [15]. CoT is useful for bigger tasks that require common-sense reasoning and can be both zero-shot (zero-CoT) and few-shot (few-CoT) [15]. Zero-CoT is less time consuming, since it does not require example data, but also less controllable than few-CoT. Kojima et al. [16] combat this dilemma through first using zero-CoT to generate outputs where the best ones are then used as data for few-CoT prompts. Park et al. [17] also reuses outputs to create prompts by prompting an LLM to generate questions about the input that it answer, where the answers help summarize information. They found it was an effective method for eliciting deeper understanding in the LLM, improving its ability to summarise, a task many models otherwise struggle with.

D. LLMs and TRPGs

There are a few types of studies looking at how LLMs can be integrated with TRPGs. In the most prevalent studies, an LLM is given a dataset of transcripts from TRPG sessions and is then tasked to predict which player is to act next and what they will do, based on the current context [6]–[9]. These studies will be referred to as *Player Action Prediction* (PAP) studies. The method used for PAP was first introduced by Louis and Sutton [18]. The subsequent studies have mainly extended on this by suggesting novel prompting methods and applied the technique to new data sets.

The datasets used consist of TRPG transcripts from play-by-post forums, where D&D is the most common TRPG used [6], [7], [9]. Callison-Burch et al. [7] labeled their transcripts if players were speaking in or out of character through training on play-by-post forums with separate chats for in and out of character conversation. This is extended upon in subsequent studies [6], [8], [9]. Zhou et al. [9] and Liang Zhu and Yang [8] include intention of character actions in their datasets, where the former includes GM intentions and the latter defines skill-checks as intentions, to improve predictions. To evaluate PAP studies, the most common metric used is that of perplexity [6], [7], which looks at the range of prediction when predicting next character and action. The lower the perplexity score the better. Some studies [6]–[8] also include subjective evaluations, using experienced D&D players, looking at groundedness (how human-like the responses are) and if there are factual errors. To compare results, ablation is used where parts of information included in the datasets are removed [7]–[9]. All studies found that including more context-based information, especially regarding character background, improved the models' predictions.

III. Problem

Previous PAP studies [2], [8], [9] contribute with interesting insights, touching upon aspects of frames, such as including in and out of character labels [6], [7] or that of trying to define the intentions of players [8], [9] to improve their results. These studies highlights that there is a need for understanding not only the rules of the game context, but that of the fantasy frame, but how this is to be done remains unanswered.

Alignments guide the players in how their character should act, and is a common place of struggle for self-players [3]. In comparison to character traits that are physical, where the player performs the actions through rolling dice, personality traits such as alignments can only be performed through role-playing and speaking as one's character. Alignments could therefore be one of the keys to teaching an AI how to role-play and understand whether other players are role-playing too. Despite their close connection to role-playing and fantasy frame, alignments have yet to be mentioned in previous studies.

To explore how the fantasy frame can be understood by AI, this paper proposes the use of alignments to see if a player is acting in- or out of character. For this, the following questions are posed:

- How can an LLM be used to detect to what extent a player is acting according to its character's alignment when playing Dungeons and Dragons?

- How can prompting be used to achieve this?

IV. Method

A. Dataset and Model

Because of GPT [12] being one of the biggest LLMs and its API allows for testing prompts without using the data for feedback training, it became the model of choice for developing and testing prompts. The model used in development and for pilot testing is GPT-3.5.

The dataset used for the prompts is a combination of the D&D alignments from the 5th edition basic rule book [19] as well as transcripts from Critical Role (CR) [20], an online series where actors play D&D live. The transcripts are written after each episode, making it based on live playing sessions, which is preferred over play-by-post transcripts, where play-sessions can happen over longer periods of time, lacking the natural flow of real-life playing [8]. The dataset includes characters of the alignments neutral good (NG), chaotic good (CG) and chaotic neutral (CN), with each alignment being represented by at least two different actors. Because of GPT knowing of CR before, all characters and settings have been anonymized.

B. Controlled Chain of Thought

In order to elicit an understanding of the alignment rules, the prompting method CCoT was developed (Figure1). Because the alignments are only described briefly in the rule book, in step 1, GPT is prompted to generate questions regarding the alignments that, when answered, would give it enough information that it could determine the alignment of a character. It is a similar method to that of Park et al. [17] with the difference that, instead of generating questions that can already be answered with the information given, GPT is asked to generate questions that it cannot answer without further information, highlighting the areas that are *not* understood by the model. In step 2, GPT is prompted with a character description taken from the CR dataset representing a specific alignment, and asked to answer the questions previously generated, citing examples from the description when doing so. This has a similar effect to CoT, where the model goes through the information step-by-step, but in a controlled manner without manually creating an example of reasoning, relying on the questions instead. Lastly, in step 3 GPT is prompted to summarise the answers elicited into a general definition of the alignment in question. Because LLMs are designed to predict the next word to come, they can sometimes focus too much on details in text, making the output too specific. Since the desire is to have GPT understand a specific alignment that can represent several characters, the summary generalises the information, generating a higher-level understanding of the alignment.

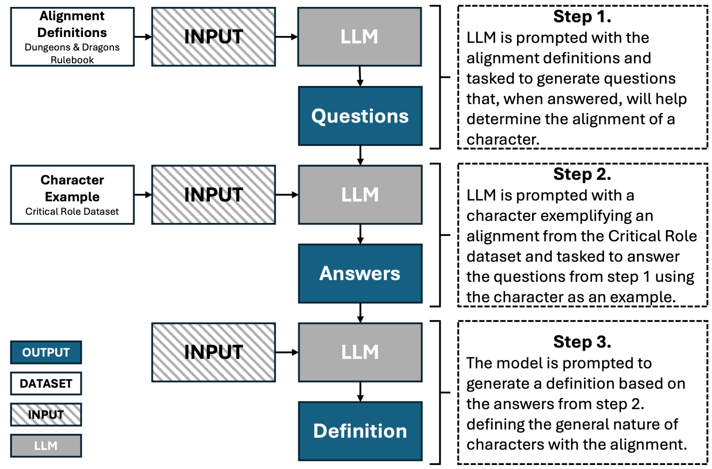

Fig. 1. Controlled Chain of Thought

C. Pilot Testing

To see if CCoT generates a higher-level understanding of alignments in GPT an initial pilot testing was held with a baseline method using no prompts (zero-shot) and CCoT were tasked to detect characters of the alignment NG. The test consisted of four unique character descriptions (explanation of a character's background) and three scenarios (excerpts of in-game conversation-logs) of different lengths consisting of 11 characters, amounting to 15 characters in total, 7 of which being of the alignment NG. This ensured inclusion of both static descriptions of the characters, as well as dynamic in-game dialogue. As metrics, the testing followed the method of Liang, Yang and Zhu [8]. They include both precision and recall to generate the harmonic mean (HM) of their results.

V. Preliminary Results

Using the baseline prompt, 5 characters were identified as NG, 2 of which being correct, misidentifying 3 non-NG characters and missing the remaining 5 of the NG characters. Using the CCoT prompts, the LLM correctly identified all 7 NG characters, and did not misidentify any others (see Table I). However, in the last and longest scenario some uncertainty was identified when using CCoT, which is not visible in the data. GPT stated that it could be possible for a non-NG character to be NG, but that there was not enough information to conclude this. This implies that current metrics fall short of fully representing the results, since these types of uncertainties is not represented in the data. Possible suggestions to solve this is by tasking GPT to identify all alignments at once, or to represent the alignment variables lawful-chaos and good-evil separately in the data.

TABLE I
ALIGNMENT PREDICTION RESULTS

Test	Precision (p)	Recall (r)	Harmonic Mean ($\frac{2pr}{p+r}$)
Baseline	0.40	0.28	0.33
CCoT	1.0	1.0	1.0

GPT-3.5 is tasked to identify neutral-good characters (n=7) in the provided excerpts with a total of 15 characters.

VI. DISCUSSION

It is the understanding of frames that is the key to role-play. Without distinguishing these frames, and noting their sensitive relationship, an AI can never understand role-play and therefore never engage in it. It is important to note that, even though this paper has highlighted the importance of frames, it is still unclear if incorporating alignments extends this frame knowledge to an AI. The aspiration is that this inclusion of frames through alignments will open up further discussing on what signifies role-play.

Having the LLM itself point out missing information through generating questions seems to elicit an understanding of what it does not understand, which can be useful for commonsense knowledge reasoning. CCoT shows promise in its ability to generate higher-level understanding of alignments in GPT, which could possibly be a first step in understanding when and how players role-play. Moving forward to larger scale testing, an ablation study will be done to evaluate the effect each prompting step has on the over-all result. However, since there is still data that is difficult to represent quantitatively, the metrics currently used should be revised. Despite the TRPG-focus of the paper, it is possible that CCoT is a prompting method that could prove useful outside of TRPG settings, because of it generating a more controlled CoT without requiring the user to manually create or find prompt examples.

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Appendix J - Vocabulary List

Ablation	Evaluation technique commonly used when evaluation an LLM's architecture, its training data or the structure of prompts to see what affects its performance.
Alignments	A roleplaying mechanic used in Dungeons and Dragons to define a character's view on social norms and morality.
Artificial Intelligence	A computer system which simulates human intelligence processes.
Commonsense Reality Frame	A frame defined by Fine, which represents context outside of a TRPG and all information within it.
Content Generation	Studies focusing on using LLMs to generate content in a TRPG setting.
Control Chain of Thought	A prompting method where an LLM is prompted to generate a reasoning chain to break down larger tasks into smaller ones.
Decoder	A part of the transformer structure which is trained to generate text by making predictions of the next word to come, based on the context/prompt given.
Encoder	A part of the transformer structure which is trained to encode words as tokens and storing them inside vectors, positioning them based on their semantic similarity.
Fantasy Frame	A frame defined by Fine, which represents the fantasy context of a TRPG where players become their character through roleplay.
Few-shot	A prompting technique which includes one or more examples depicting the desired behaviour of an LLM.
Game Context Frame	A frame defined by Fine, which represents all information surrounding a TRPG such as rules, dice roll, any conversation made regarding game rules.
Game Master	The player in a Tabletop roleplaying session who serves as a referee by keeping track of the rules, while simultaneously guiding the players through a world and storyline they have created.
Game Master Assistance	Studies focusing on using LLMs to help assist Game Masters when playing Tabletop roleplaying games.

Generative Agent	Software that uses generative models (e.g. large language models) to simulate human behaviour.
GPT	A decoder based LLM made by the company OpenAI.
Hallucinations	The occurrence of out of context information generated by an LLM, often as a consequence of lacking information or badly worded prompts.
keying	A phrase coined by Goffman explaining the act of adapting ones actions and way of speaking based on the current social context. Fine also includes adaption of awareness in his study
Large Language Model	a generative AI trained on a large corpus of text-based data to generate text. There are decoder, encoder and decoder-encoder based LLMs.
Narration Challenge	Studies using TRPG transcripts as datasets to train LLMs on narrative generation and summarisation tasks.
Parameter	The part of an LLM that stores training data in the form of tokens and vectors. More parameters equate to more training data.
Perplexity	A measurement looking at the range an LLM is making predictions when generating text, indicating its accuracy. Low perplexity means better accuracy.
Player Action Prediction	Studies focusing on training LLMs on TRPG transcripts to predict the next player in a session to act and what it will do.
Prompt Engineering	A field in LLM research focused on development and testing of prompts the best outputs.
Prompts	Inputs made by users to elicit a specific behaviour or output form an LLM
Reasoning	A break-down of an LLMs calculations in predicting the next word to generate often elicited through CoT prompting. Not to be confused with human reasoning.
Roleplaying Frames	The frames defined by Fine consisting of commonsense reality, game context and fantasy.
Tabletop Roleplaying Games	Roleplaying games usually played in groups of players, with one serving as a game master, with the goal of immersion into a fantasy setting.
Token	A unit representing the smallest piece of information of semantic meaning in a word (sometimes part of a word).

Zero-shot	A prompting method where the user only inputs the desired behaviour or output without examples.